ENGLISH

Belli - war/conflict
Anthrop - human
Amphi - Dual / both

Choose the correct SYNONYM for the following

1) Ameliorate
   a) Understand b) Eliminate
   c) Improve d) Confined
2) Abysmal
   a) Bottomless b) Addition
   c) Fatalist d) Valuable
3) Blandishment
   a) Punishment b) Cheating
   c) Unwilling d) Flattery
4) Brochure
   a) News b) Magazine
   c) Pamphlet d) Annals
5) Circumvent
   a) Caution b) Frustrate
   c) Diversity d) Quote

Choose the correct ANTONYM for the following

6) Acrimonious
   a) Bitter b) Soothing
   c) Plain d) Intelligent
7) Archetype
   a) Copy b) Generous
   c) Origin d) Factory
8) Belligerent
   a) Loud b) Interested
   c) Peaceful d) Popular
9) Bustle
   a) Ado b) Rustle
   c) Stir d) Quiet
10) Cajole
   a) Synthesize b) Antagonize
   c) Coax d) Proper

Point out which underlined part of the following sentence is incorrect.

11) There (A) have (B) been heavy (C) rainfall yesterday (D).
12) Aftab (A) as well as his (B) class fellows are (C) playing (D) hockey.
13) Neither the (A) head constable nor (B) other (C) policemen is (D) injured.
14) Every (A) leaf and every flower proclaim (B) the (C) glory (D) of God.
15) Six miles (A) are (B) a (C) long (D) distance.

Fill in the following blanks with the most appropriate word.

16) ________ should not hate the poor.

a) The rich b) Rich
c) A rich d) An rich

17) We take __________ at 8 am daily.
a) the breakfast b) A breakfast
c) breakfast d) none of these

18) The more you work, __________ it is.
a) better b) the better
c) a better d) more better

19) ____ Himalayas lie in the north of Pakistan
a) A b) The
c) An d) none

20) Today we need __________ to advocate for us.
a) the Quaid-e-Azam b) Quaid-e-Azam
c) a Quaid-e-Azam d) none of these

Read the following comprehensions and answer the questions that follow

PASSAGE 1

In Asia and much of the Third World, trees are still destroyed in the old fashioned way: they are cut down for fuel and cropland. In Europe, there is new and potentially more deadly culprit. The Germans call it 'Waldsterben', the dying forest syndrome. But the disease is far more than a German phenomenon. Since it was first observed by German scientists in the autumn of 1980, the mysterious malady has raced across Europe, blighting woods in countries as far apart as Sweden and Italy. Explanations for the epidemic range from a cyclic change in the environment to a baffling form of tree cancer. But the most convincing evidence points to air pollution. Indeed, saving the rapidly deteriorating Forests of Europe will probably require a two-pronged strategy: an offensive campaign that includes the breeding of pollution- immune trees and a defensive scheme that calls for reductions in toxic emissions. But both will require more money than is currently being spent on such measures, as well as total commitment to protecting the environment.

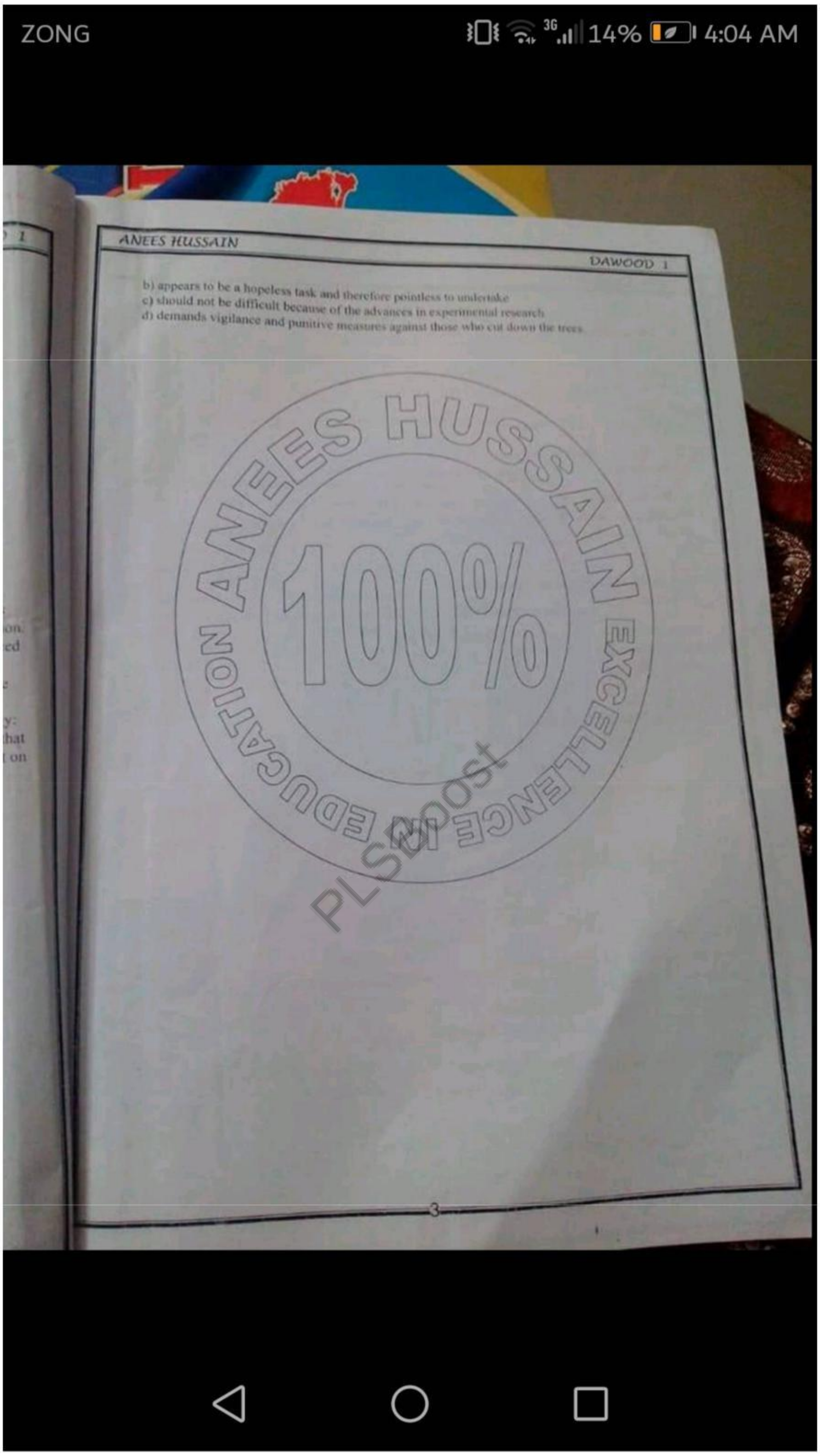
21) According to this passage, which one of the following statements is correct"?
a) There is less damage in Asia than in Europe
b) More forests are dying in Germany than anywhere else in Europe
c) A cyclic change in the environment is responsible for deforestation
d) Air pollution is the main culprit of destroying European forests

22) The dying forest syndrome is a disease That
a) is peculiar to the forests of Asia
b) has spread rapidly over the forests of Europe
c) is confined to the forests of Germany
d) has affected forests all over me world

23) The writer suggests that
a) it is no longer possible to grow trees in industrialized areas
b) pollution immune trees will absorb toxic emissions
c) all pollution-prone trees should be destroyed
d) it is not possible to grow trees that remain unaffected by pollution

24) The writer's approach toward the problem of forest devastation is one of
a) tolerance b) indifference
c) well thought—out strategy d) despondency

25) Saving the trees of European forests
a) requires a much bigger budget


ANEES HUSSAIN

DAWOOD 1

b) appears to be a hopeless task and therefore pointless to undertake
c) should not be difficult because of the advances in experimental research
d) demands vigilance and punitive measures against those who cut down the trees

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ANEES HUSSAIN Dawood 1

CHEMISTRY

1. One atm is equal to ________ Pascal
   a.) 760 b.) 101325
   c.) 14.7 d.) 1.01325
2. If $V_1 = 5$ liters, $P_1 = 2$ atm, $T_1 = 273\,^\circ C$, $T_2 = 0\,^\circ C$ and $V_2 = ?$ when $P_2 = 1$ atm
   a.) 5 liters b.) 2.5 Liters
   c.) 12.5 Liters d.) 10 Liters
3. ________ lies in U.V region.
   a.) Balmer Series b.) Lyman Series
   c.) Paschen Series d.) All of the above
4. The maximum number of electrons that can be present in a main energy level is
   a.) $2n^2$ b.) 2n
   c.) 2 d.) n
5. The two electrons in same orbital should have:
   a.) Similar spin b.) opposite spin
   c.) parallel spin d.) Perpendicular spin
6. Mass of electron is:
   a.) 9.1×10^{-31} Kg b.) $19 \times 10^{[illegible]}$ Kg
   c.) 9×10^{-30} Kg d.) $9.1 \times 10^{[illegible]}$ Kg
7. Change of Enthalpy is denoted
   a.) H b.) ΔH
   c.) E d.) ΔE
8. Rate constant for gaseous reaction is written as:
   a.) K_c b.) K_p
   c.) K_n d.) K_{sp}
9. Oxidation state of oxygen in per-oxides
   a.) 2 b.) -1
   c.) -2 d.) 1
10. Rate of reaction:
    a.) Increases as the reaction proceeds
    b.) Decreases as the reaction proceeds
    c.) Remain same as the reaction proceeds
    d) May increase or decrease as the reaction proceeds
11. Photochemical reaction usually have the order
    a.) One b.) Two
    c.) Three d.) Zero
12. ________ are biocatalysts.
    a.) Organic acids b.) Organic bases

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c.) Enzymes d.) All

13. Metalloids gives ________ oxides.
    - a.) Metal
    - b.) Amphoteric
    - c.) Homogeneous
    - d.) Heterogeneous
14. Fifth period of modern periodic table contains:
    - a.) 2
    - b.) 6
    - c.) 12
    - d.) 18
15. Electro negativity difference for ionic bond must be greater than
    - a.) 1.6
    - b.) 1.7
    - c.) 1.8
    - d.) 1.9
16. The Measurement of degree of disorder
    - a.) Entropy
    - b.) Enthalpy
    - c.) Rate of Reaction
    - d.) Temperature
17. The unit of Kc for reaction $N_2 + O_2 \rightleftharpoons 2NO$?
    - a.) Mol dm^{-3}
    - b.) Mol dm^{3}
    - c.) $mol^{2} dm^{6}$
    - d.) None
18. If two lone pairs are present then bond angle of tetrahedral compound reduce to ________
    - a.) 109.5°
    - b.) 107.5°
    - c.) 104.5°
    - d.) None
19. ________ is the formula of fluorspar?
    - a.) Na_3AlF_6
    - b.) CaF_2
    - c.) $CaF_2.3Ca$
    - d.) AlF_3
20. HF is:
    - a.) Oxidizing agent
    - b.) Reducing agent
    - c.) Weak acid
    - d.) Strong acid
21. ________ is called oil of vitriol.
    - a.) H_2SO_4
    - b.) HNO_3
    - c.) HCl
    - d.) CH_3COOH
22. Nobel metals are dissolved in:
    - a.) Conc. H_2SO_4
    - b.) Aqua Regia
    - c.) Conc. HCl
    - d.) All of these
23. Catenation is the ability of carbon to grow in ________
    - a.) Size
    - b.) Number
    - c.) Line
    - d.) Long Chains
24. Which is the most volatile compound?
    - a.) HI
    - b.) HCl
    - c.) HBr
    - d.) HF

PLS Boost

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25. Di-ethyl ketone methyl n- propyl ketone are an example of

a.) Chain Isomerism
b.) Positional Isomerism
c.) Metamerism Isomerism
d.) Functional group Isomerism

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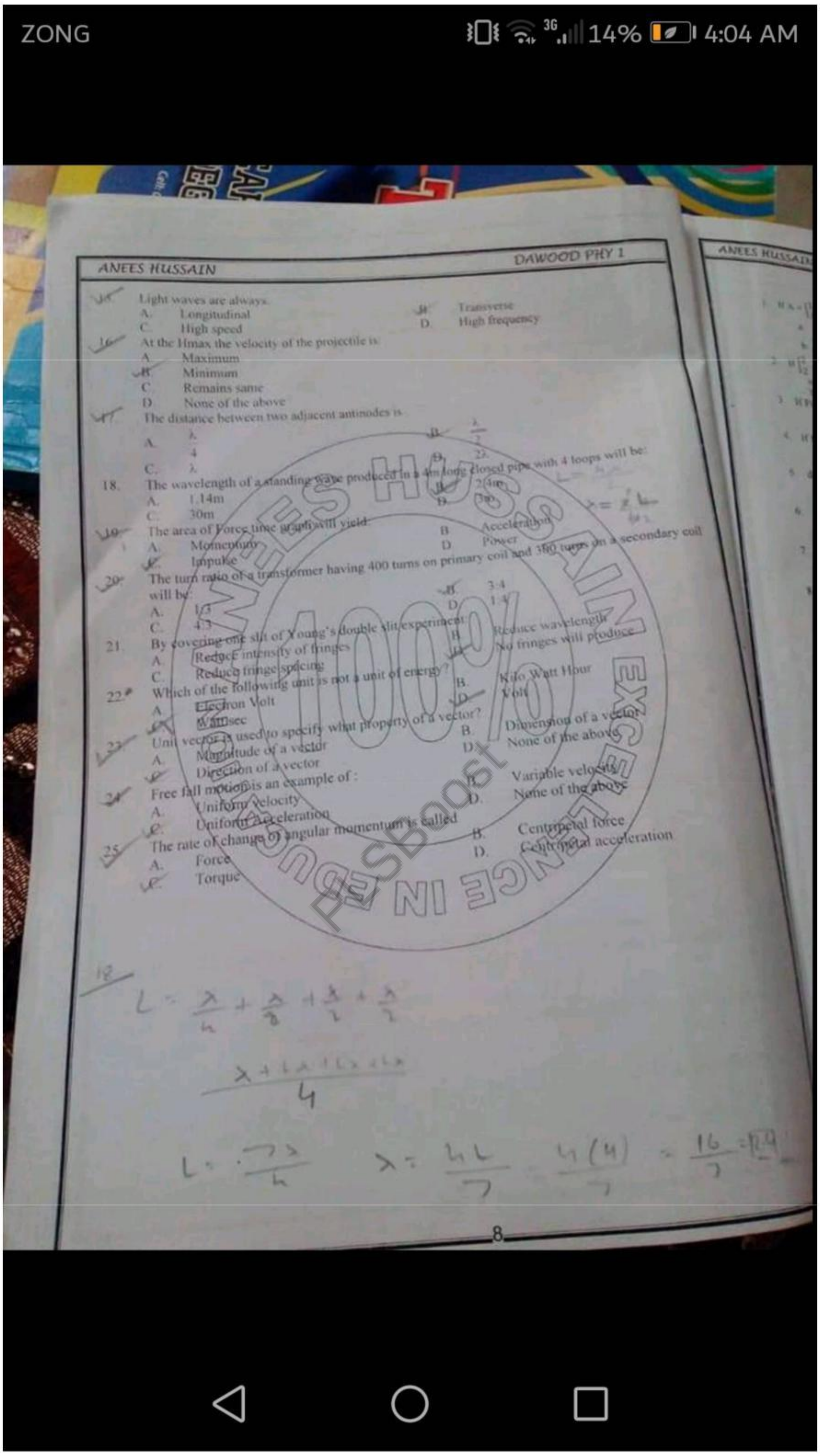
ANEES HUSSAIN DAWOOD PHY 1

PHYSICS

1. The speed of sound wave travelling through air doesn't depend upon
   A. air temperature B. atm pressure
   C. gasses in air D. thermal conductivity
2. A 2Ω resistor is connected for 0.5 min to a P.D of 1K.V. the heat developed in resistor is
   A. 36 kJ B. 36 K J
   C. 60 kJ D. 360 K J
3. Electric flux is the
   A. scalar product of the electric field and the surface area
   B. flow of electric lines per unit area
   C. scalar quantity
   D. all of them
4. Which is more penetrating?
   A. α - rays B. β - rays
   C. γ - rays D. X rays
5. An object is placed at 30 cm from a lens of focal length 15cm its image size will be?
   A. 6 cm B. 15 cm
   C. 8 cm D. 30 cm
6. Time period of simple pendulum is
   A. $T = 2\pi\sqrt{\frac{l}{g}}$ B. $\frac{1}{2\pi}\sqrt{\frac{l}{g}}$
   C. $T = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$ D. $2\pi\sqrt{\frac{g}{l}}$
7. In 1932 James Chadwick discovered
   A. Electron B. Proton
   C. Neutron D. None
8. Controlled fission reaction takes place in
   A. Cyclotron B. Nuclear reactor
   C. Sun D. Earth
9. Center of Newton's Ring is :
   A. Dark B. Both
   C. Bright D. None
10. Find the value of 'a' if two vectors $A = 2i + aj + k$ and $B = 4i\ 3j\ 7k$ are perpendicular to each other.
   A. 20 B. [illegible]
   C. 10 D. 5
11. The rate of change of angular momentum is equal to the
   A. torque B. Mass
   C. force D. energy
12. Find the work done on the body which starts from rest and moving when a force of 10N is applied on it for 4s if its mass is found to be 4Kg.
   A. 1.0 J B. 200 J
   C. 25J D. 300J

   1400 J
13. A ball is the height of 80m. Find the time taken by ball to reach ground.
   A. 16 sec B. 4 sec
   C. 32 sec D. 8 sec
14. Unit of radioactivity is.
   A. Kg/s B. Decay.sec
   C. Becquerel D. None



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ANEES HUSSAIN DAWOOD PHY 1

15. Light waves are always
A. Longitudinal
B. Transverse
C. High speed
D. High frequency

16. At the Hmax the velocity of the projectile is
A. Maximum
B. Minimum
C. Remains same
D. None of the above

17. The distance between two adjacent antinodes is
A. $\frac{\lambda}{4}$
B. $\frac{\lambda}{2}$
C. λ
D. 2λ

18. The wavelength of a standing wave produced in a 4m long closed pipe with 4 loops will be:
A. 1.14m
B. 2.4m
C. 30m
D. 3m

19. The area of Force time graph will yield
A. Momentum
B. Acceleration
C. Impulse
D. Power

20. The turn ratio of a transformer having 400 turns on primary coil and 300 turns on a secondary coil will be:
A. 1:3
B. 3:4
C. 4:3
D. 1:4

21. By covering one slit of Young's double slit experiment
A. Reduce intensity of fringes
B. Reduce wavelength
C. Reduce fringe spacing
D. No fringes will produce

22. Which of the following unit is not a unit of energy?
A. Electron Volt
B. Kilo Watt Hour
C. Watt-sec
D. Volt

23. Unit vector is used to specify what property of a vector?
A. Magnitude of a vector
B. Dimension of a vector
C. Direction of a vector
D. None of the above

24. Free fall motion is an example of :
A. Uniform velocity
B. Variable velocity
C. Uniform Acceleration
D. None of the above

25. The rate of change of angular momentum is called
A. Force
B. Centripetal force
C. Torque
D. Centripetal acceleration

18

$L = \frac{\lambda}{4} + \frac{\lambda}{2} + \frac{\lambda}{2} + \frac{\lambda}{2}$

$\frac{\lambda + 2\lambda + 2\lambda + 2\lambda}{4}$

$L = \frac{7\lambda}{4}$ $\lambda = \frac{4L}{7}$ $\frac{4(4)}{7} = \frac{16}{7} = 2.29$

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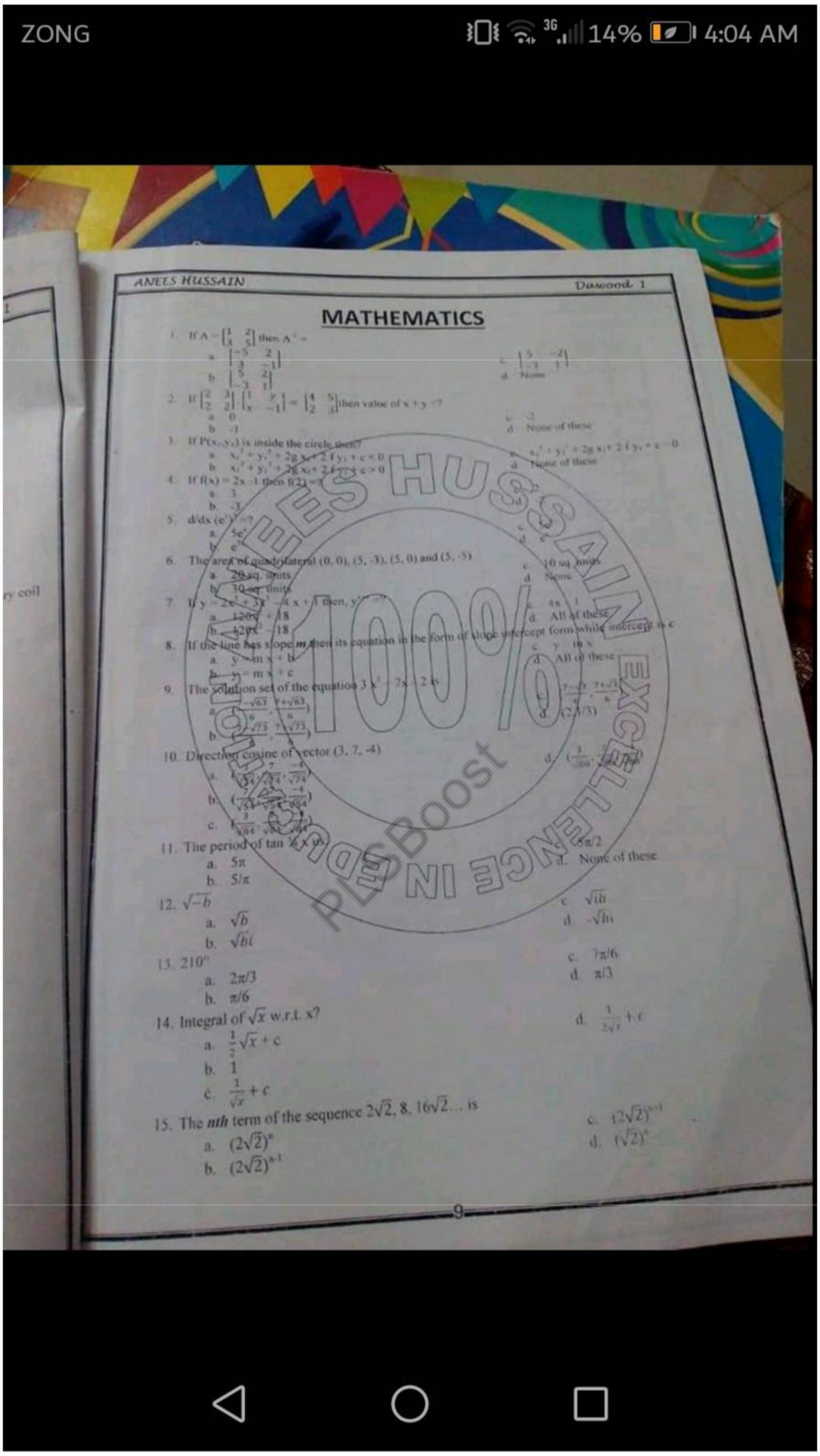
PLS Boost



ANEES HUSSAIN — Dawood 1

MATHEMATICS

1. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ then $A^{-1} =$
   a. $\begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$
   b. $\begin{bmatrix} 5 & 2 \\ -3 & 1 \end{bmatrix}$
   c. $\begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix}$
   d. None
2. If $\begin{bmatrix} 2 & 3 \\ 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & y \\ x & -1 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 2 & 3 \end{bmatrix}$ then value of $x + y$ =?
   a. 0
   b. -1
   c. -2
   d. None of these
3. If $P(x_1, y_1)$ is inside the circle then?
   a. $x_1^2 + y_1^2 + 2g x_1 + 2f y_1 + c < 0$
   b. $x_1^2 + y_1^2 + 2g x_1 + 2f y_1 + c > 0$
   c. $x_1^2 + y_1^2 + 2g x_1 + 2f y_1 + c = 0$
   d. None of these
4. If $f(x) = 2x - 1$ then $f(2)$ =?
   a. 3
   b. -3
   c. [illegible]
   d. [illegible]
5. $d/dx\,(e^x)^5$ =?
   a. $5e^x$
   b. e^{5x}
   c. [illegible]
   d. [illegible]
6. The area of quadrilateral (0, 0), (5, -3), (5, 0) and (5, -5)
   a. 20 sq. units
   b. 30 sq. units
   c. 10 sq. units
   d. None
7. If $y = 2x^5 + 3x^3 - 4x + 1$ then, y'' =?
   a. $120x^3 + 18$
   b. $120x^2 - 18$
   c. $4x - 1$
   d. All of these
8. If the line has slope m then its equation in the form of slope intercept form while intercept is c
   a. $y = mx + b$
   b. $y = mx + c$
   c. $y = mx$
   d. All of these
9. The solution set of the equation $3x^2 - 7x - 2$ is
   a. $(\frac{7-\sqrt{63}}{6}, \frac{7+\sqrt{63}}{6})$
   b. $(\frac{7-\sqrt{73}}{6}, \frac{7+\sqrt{73}}{6})$
   c. $(\frac{7-\sqrt{3}}{6}, \frac{7+\sqrt{3}}{6})$
   d. (2, 1/3)
10. Direction cosine of vector (3, 7, -4)
    a. $(\frac{3}{\sqrt{74}}, \frac{7}{\sqrt{74}}, \frac{-4}{\sqrt{74}})$
    b. $(\frac{7}{\sqrt{54}}, \frac{7}{\sqrt{54}}, \frac{-4}{\sqrt{64}})$
    c. $(\frac{3}{\sqrt{84}}, \frac{7}{\sqrt{84}}, \frac{-4}{\sqrt{84}})$
    d. $(\frac{3}{\sqrt{64}}, \frac{7}{\sqrt{64}}, \frac{-4}{\sqrt{64}})$
11. The period of tan [illegible]
    a. 5π
    b. $5/\pi$
    c. $5\pi/2$
    d. None of these
12. $\sqrt{-b}$
    a. $\sqrt{b}$
    b. $\sqrt{bi}$
    c. $\sqrt{ib}$
    d. $-\sqrt{bi}$
13. 210°
    a. $2\pi/3$
    b. $\pi/6$
    c. $7\pi/6$
    d. $\pi/3$
14. Integral of $\sqrt{x}$ w.r.t. x?
    a. $\frac{1}{2}\sqrt{x} + c$
    b. 1
    c. $\frac{1}{\sqrt{x}} + c$
    d. $\frac{1}{2\sqrt{x}} + c$
15. The *nth* term of the sequence $2\sqrt{2}, 8, 16\sqrt{2}\ldots$ is
    a. $(2\sqrt{2})^n$
    b. $(2\sqrt{2})^{n-1}$
    c. $(2\sqrt{2})^{n+1}$
    d. $(\sqrt{2})^n$

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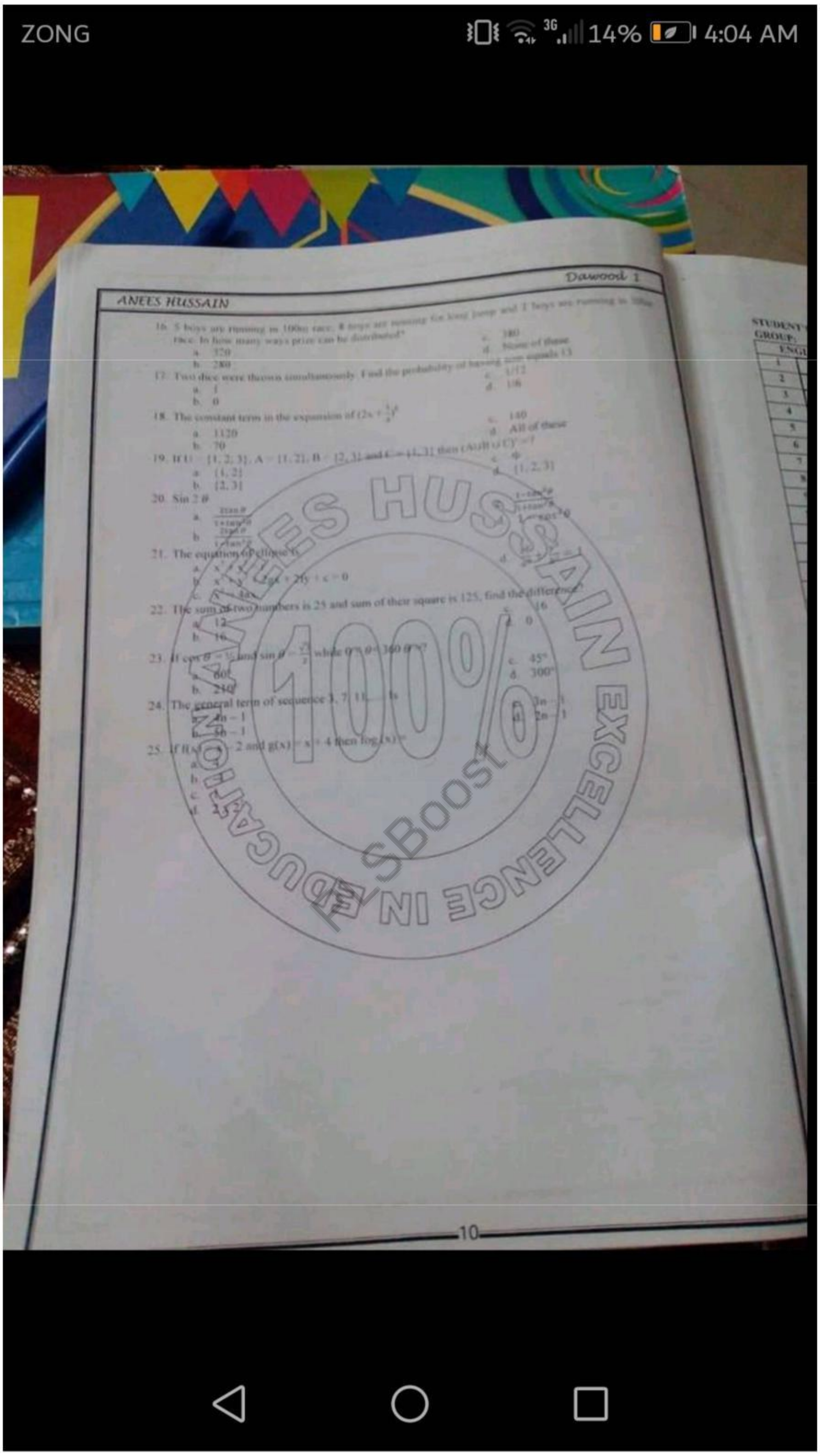


ANEES HUSSAIN

Dawood 1

16. 5 boys are running in 100m race, 8 boys are running for long jump and 3 boys are running in 200m race. In how many ways prize can be distributed?
a. 320
b. 280
c. 380
d. None of these

17. Two dice were thrown simultaneously. Find the probability of having sum equals 13
a. 1
b. 0
c. 1/12
d. 1/6

18. The constant term in the expansion of $(2x+\frac{1}{x})^8$
a. 1120
b. 70
c. 140
d. All of these

19. If U = {1, 2, 3}, A = {1, 2}, B = {2, 3} and C = {1, 3} then $(A \cup B \cup C)'$ = ?
a. {1, 2}
b. {2, 3}
c. Φ
d. {1, 2, 3}

20. Sin 2θ
a. $\frac{2\tan\theta}{1+\tan^2\theta}$
b. $\frac{2\tan\theta}{1-\tan^2\theta}$
c. $\frac{1-\tan^2\theta}{1+\tan^2\theta}$
d. [illegible]

21. The equation of ellipse is
a. [illegible]
b. $x^2+y^2+2gx+2fy+c=0$
c. $y^2=4ax$
d. [illegible]

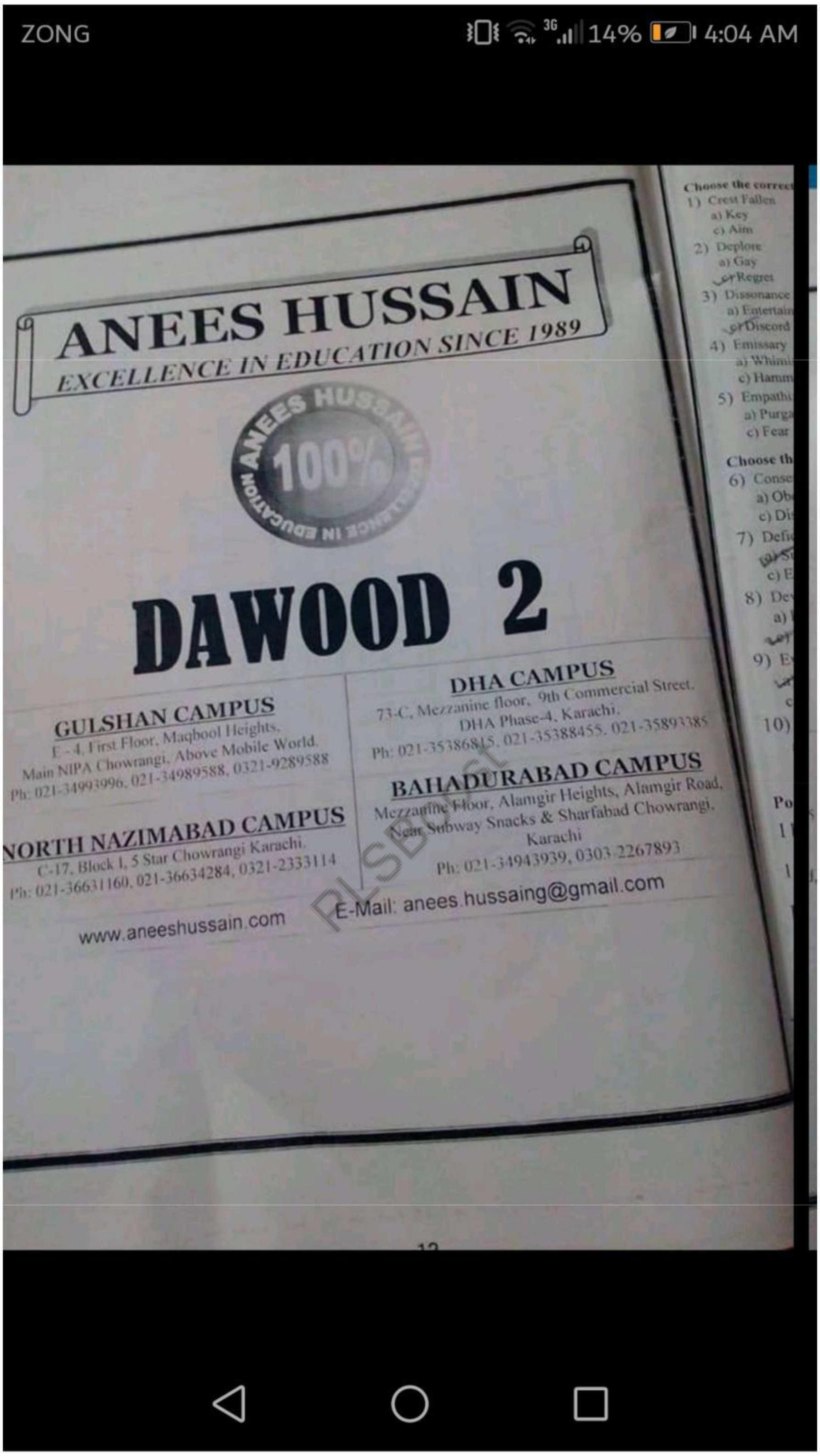
22. The sum of two numbers is 25 and sum of their square is 125, find the difference?
a. 12
b. 16
c. 16
d. 0

23. If cos θ = ½ and sin θ = $\frac{\sqrt{3}}{2}$ while 0° ≤ θ ≤ 360, θ = ?
a. 60°
b. 210°
c. 45°
d. 300°

24. The general term of sequence 3, 7, 11, ... is
a. 4n − 1
b. 5n − 1
c. 3n − 1
d. 2n − 1

25. If f(x) = x − 2 and g(x) = x + 4 then fog(x) = ?
a. [illegible]
b. [illegible]
c. [illegible]
d. 2.5 [illegible]

10



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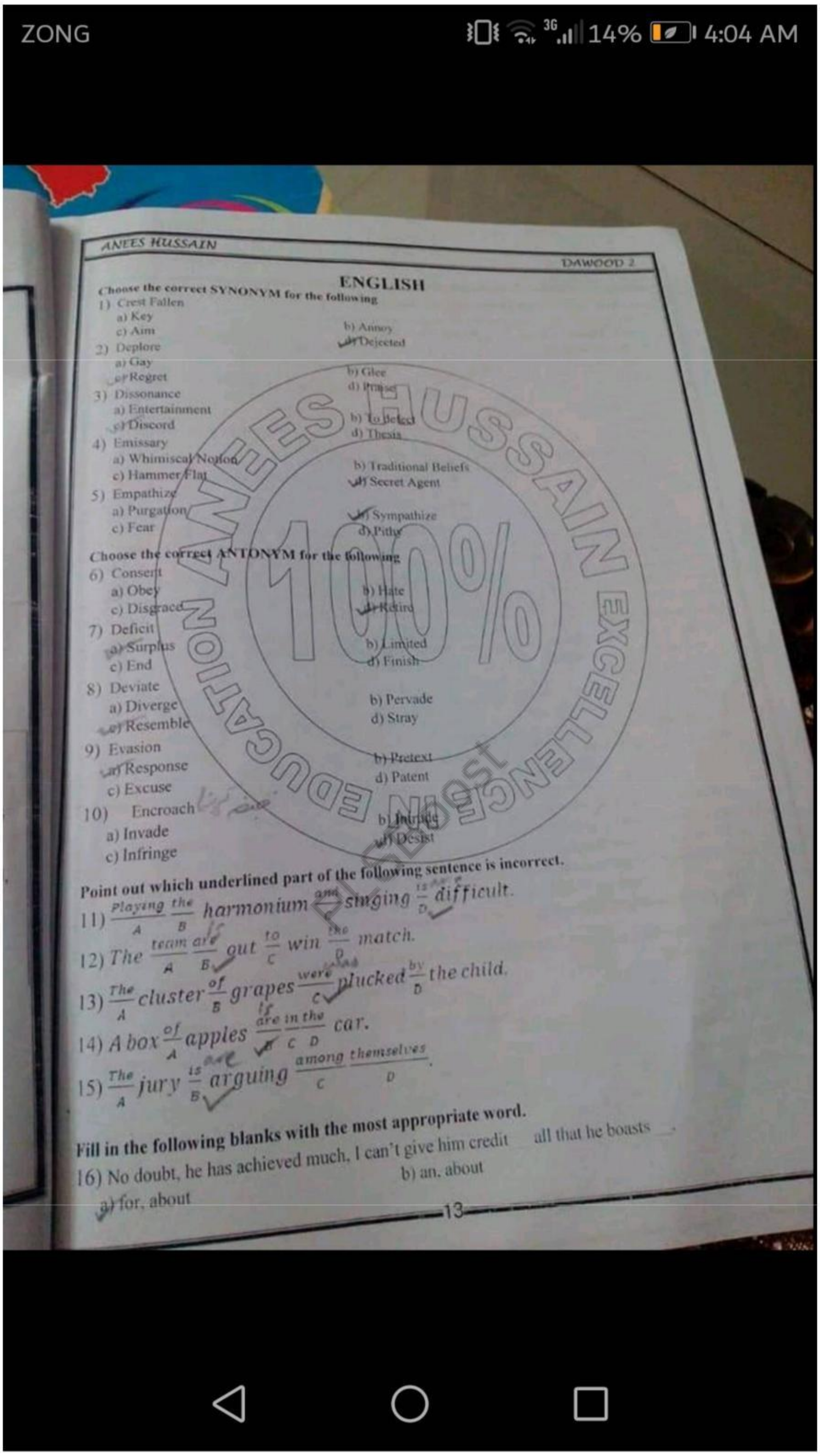

ANEES HUSSAIN

DAWOOD 2

ENGLISH

Choose the correct SYNONYM for the following

1) Crest Fallen
   a) Key
   b) Annoy
   c) Aim
   d) Dejected
2) Deplore
   a) Gay
   b) Glee
   c) Regret
   d) Praise
3) Dissonance
   a) Entertainment
   b) To detect
   c) Discord
   d) Thesis
4) Emissary
   a) Whimsical Notion
   b) Traditional Beliefs
   c) Hammer Flat
   d) Secret Agent
5) Empathize
   a) Purgation
   b) Sympathize
   c) Fear
   d) Pithy

Choose the correct ANTONYM for the following

6) Consent
   a) Obey
   b) Hate
   c) Disgrace
   d) Retire
7) Deficit
   a) Surplus
   b) Limited
   c) End
   d) Finish
8) Deviate
   a) Diverge
   b) Pervade
   c) Resemble
   d) Stray
9) Evasion
   a) Response
   b) Pretext
   c) Excuse
   d) Patent
10) Encroach
   a) Invade
   b) Intrude
   c) Infringe
   d) Desist

Point out which underlined part of the following sentence is incorrect.

11) Playing (A) the (B) harmonium and (C) singing is (D) difficult.
12) The team (A) are (B) out to (C) win the (D) match.
13) The (A) cluster of (B) grapes were (C) plucked by (D) the child.
14) A box of (A) apples are (B) in the (C D) car.
15) The (A) jury is (B) arguing among (C) themselves (D).

Fill in the following blanks with the most appropriate word.

16) No doubt, he has achieved much, I can't give him credit ___ all that he boasts ___.
   a) for, about
   b) an, about

13

c) for, to d) about, the

17) By wearing contact lenses that corrected his colour blindness, he was ______ for the job.
a) was able for qualifying b) was able to qualify
c) was ably qualified d) was ably qualifying

18) These terms are very harsh; I doubt the CEO will agree ______ them.
a) for b) with
c) of d) to

19) I'm afraid that I have to agree ______ your opinion about the new actuary.
a) with b) over
c) from d) of

20) ______ only chance you are left with is to re-appear for ______ paper.
a) X, the b) The, the
c) The, X d) The, a

PASSAGE

For fourteen and half months I lived in my little cell or room in Dehradun jail, and I began to feel as if I was almost a part of it. I was familiar with every bit of it. I knew every mark and dent on the whitewashed walls and on the uneven floors and on the ceiling with the moth eaten rafters. In the little yard outside I greeted little tufts of grass and odd bits of stone as old friends. I was not alone in my cell, for several colonies of wasp and hornets lived there, and many lizards found the home behind the rafters, emerging in the evening in the search of prey.

21) Which of the following explains best the sentence in the passage "I was almost a part of it"?
a) I was not alone in the cell
b) I was familiar with every bit of the cell
c) I greeted little tufts of grass like old friends
d) I felt quite at home in the cell

22) The author of the passage seems to suggest that
a) It is possible to adjust one-self to uncongenial surroundings.
b) The conditions in Indian prisons are not bad
c) It is not difficult to spend one's time in prison
d) There is a need to improve the conditions in our jails

23) The passage attempts to describe:
a) The general conditions of the country's jail
b) The prisoner's conscious efforts to overcome the loneliness
c) The prisoner's capacity to notice the minute details of his surroundings
d) The prisoner's ability to live happily with other creatures

Choose the option which is closely related to the capitalized word sentence.

24) Book : Cupboard
a) Glass : Drink b) Cart : City
c) Waste : Jug d) Wine : Bottle

25) Circle : Square
a) Oval : Rectangle b) Ball : Batt
c) Sphere : Globe d) Volume : Area

ANEES HUSSAIN

Dawood 2

CHEMISTRY

1. If the density of substance is 3g/cm³ and mass is 60 then what will be the volume
   a. 20 cm³
   b. 40cm³
   c. 60cm³
   d. 180cm³
2. 20°C is equal to how many °F
   a. 0.9
   b. 68
   c. 6.7
   d. 4
3. Which of the following pair is not covalent
   a. CH_4, CCl_4
   b. MgO, NO_2
   c. CO_2, Al_2O_3
   d. NaCl
4. Internal resistance to flow of liquid
   a. Viscosity
   b. Surface tension
   c. Vapour pressure
   d. Evaporation
5. Mass of 10 mole of carbon
   a. 0.12 kg
   b. 0.120 kg
   c. 12 kg
   d. 1.2 kg
6. Which of the following has paramagnetic properties
   a. Cu
   b. Na
   c. Ni
   d. K
7. Which one of the compound is used as raw material for urea
   a. Ammonium acetate
   b. Ammonia
   c. Both a and b
   d. None of these
8. Which one of the following will noteffect the rate of reaction
   a. Temperature
   b. Concentration
   c. Catalyst
   d. Material of Container
9. Polyethylene is a
   a. Polymers
   b. Ester
   c. Synthetic fiber
   d. None of these
10. Aqueous solution of NaCl on electrolysis gives
    a. Sodium at anode
    b. Chloride to cathode
    c. Hydrogen to cathode
    d. Hydrogen gas at anode
11. Hydrogen atom consist of
    a. One electron one neutron
    b. One electron one protron
    c. One protron one neutron
    d. One electron, one protron, and one neutron
12. Which of the following are isomorphic pair
    a. SiO_2, $AgNO_3$
    b. $CaCO_3$, Na_2O
    c. $NaNO_3$, $CaCO_3$
    d. NaF, MgO
    e. Both c and d
13. Maximum electron in s block are
    a. Four
    b. Three
    c. Two
    d. One

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Dawood 2

14. The direction and extend of chemical reaction are told by ______ constant
   a. Rate
   b. Equilibrium
   c. Catalyst
   d. None of these
15. The unit of dipole moment is
   a. Debye
   b. Poles
   c. Poise
   d. None of these
16. Phenol forms phenoxides of the correspond metal with an
   a. Metal
   b. Acid
   c. Alkali
   d. Carbon compound
17. Phenol is
   a. Colorless solid
   b. Poisonous gas
   c. Non Poisonous gas
   d. Amorphous solid
18. Hybridization of phenol is
   a. Sp^3
   b. Sp^2
   c. Sp
   d. None of these
19. Carbon form
   a. metallic bond
   b. covalent
   c. Ionic bond
   d. Polar ionic bond
20. Williamson synthesis is used to prepared
   a. Acetone
   b. Diethyl ether
   c. PVC
   d. Bakelite
21. Ether are isomers with
   a. Alcohol
   b. Aldehyde
   c. Ketones
   d. None of these
22. Diethyl ether is used as
   a. Antibiotic
   b. Antiseptic
   c. Anesthesia
   d. Analgesic
23. Marsh gas mainly contain
   a. CH_4
   b. C_2H_6
   c. CO
   d. CO_2
24. Which has least C–C bond length
   a. Ethane
   b. Ethene
   c. Ethyne
   d. Benzene
25. Hybridization of olefin is
   a. Sp^3
   b. Sp^2
   c. Sp
   d. None of these

$122 = \frac{5}{9} Tc + 32$

$1098 - 32 = 5Tc$

1066

5

122×9

1098

$- 32$

1066

PHYSICS

1. The motion of charge particle controlled by
   A. electric current B. electric power
   C. electric & magnitude field D. gravitational field
2. After a complete cycle of a carnot engine the internal energy
   A. increases B. decreases
   C. remain same D. none
3. Transistor is used most as
   A. Amplifier B. Moderator
   C. Switch D. All Of the Above
4. A projectile is launched for a maximum range of 400m, which of the following co-ordinate will indicate its position at Hmax?
   A. (200,200) B. (100,100)
   C. (200,100) D. (100,200)
5. A force of 20N is applied on two springs connected in series combination and each spring is of 2N/m stiffness. If a mass is attached with them the deflection produce by it will be
   A. 1m B. 1.5m
   C. 2m D. None
6. What will be the centripetal force if the velocity is doubled and radius is increase by 4:
   A. 4 times B. Remain same
   C. Doubled D. ½ times
7. Precondition for lazing action in photodiode is to
   A. Enlarge the depletion region B. Achieve population inversion
   C. Reduce the band gaps D. Start spontaneous emission
8. A galvanometer is converted into an ammeter by connecting with it
   A. A low resistance in parallel B. A low resistance in series
   C. A high resistance in parallel D. A high resistance in series
9. Ten resistors each of 10 Ω are connected in parallel. The net resistance is
   A. 10 Ω B. 0.1 Ω
   C. 1 Ω D. 0.001 Ω
10. 122° Fahrenheit temperature is equal to
    A. 50° C B. 90° C
    C. 323°C D. 363° C
11. What will be the decay constant of a radioactive material having half life of 1600yrs?
    A. $2.3\times 10^{-11}/sec$ B. $5.3\times 10^{-11}/sec$
    C. $1.3\times 10^{-11}/sec$ D. None
12. A convex lens of 20 cm. focal length is used to form an erect image which is twice as large as the object. The position of object from the lens is
    A. 5 cm B. 10 cm
    C. 115 cm D. 20 cm
13. Colour of light is determined by.
    A. Velocity in Air B. Amplitude of vibration
    C. Wave length D. Frequency
14. By cutting a lens vertically its focal length of remaining pieces will:
    A. Increases B. Remains same
    C. Decrease D. None

15. Physical quantity tends to rotate a body is ______
A. Force
B. Momentum
C. Energy
D. Torque

16. What will be the length of the day when the radius of Earth is halved but mass remains same
A. 12Hrs
B. 6Hrs
C. 24Hrs
D. 8Hrs

17. Transistor is used to amplify
A. voltage
B. current
C. both
D. none of the above

18. Ten equal resistors, each of resistance 10Ω is connected in parallel. The equivalent resistance of the circuit is
A. 1 Ω
B. 10 Ω
C. 100 Ω
D. 50 Ω

19. A photon is moving with speed of light, its rest mass will be
A. Equal to electron
B. Equal to proton
C. Equal to Carbon
D. Zero

20. A train of 150m in length is going towards north at a speed of 10m/s. A parrot flies at a speed of 5m/s towards south parallel to the railway track. The time taken by the parrot to cross the train is equal to
A. 20 s
B. 10 s
C. 40 s
D. 50 s

21. Unit of static Coefficient of friction is ______
A. poise
B. Newton
C. dyne
D. none

22. The relative intensity (I/I_o) of the sound of a jet engine is 10^{13}. The intensity level in dB will be
A. 13
B. 130
C. 1300
D. 1.3

23. The point where the amplitude is zero in a standing wave pattern is called
A. Waves number
B. Anti Anode
C. Node
D. None of the above

24. A bomb is dropped from an airplane moving with a horizontal velocity of 200Km/h. if air resistance is neglected and stone take 5s to reach the ground, find the altitude of the plane:
A. 120m
B. 200m
C. 125m
D. 400m

25. The work produced when a force of 10N displaces an object from position (1,2,1) to (2,3,1) will be:
A. 10J
B. 14J
C. 18J
D. 3J

24
$v_i = 0$
$a = 10$
$h = ?$
$t = 5\,sec$
$S = v_i t + \frac{1}{2}at^2$
$S = \frac{1}{2}(10)(25)$
$\boxed{S = 125\,m}$

$10 \times (1, 1, 0)$
$10 \times \hat{i} + \hat{j}$
$10 \times \sqrt{2}$
$\boxed{14J}$

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MATHEMATICS

(1) Let M = {r, s, t}

i. $r \in M$
ii. $r \subset M$
iii. $\{r\} \in M$
iv. $M \supset r$

a. i only
b. iii only
c. i & iv
d. i & iii
e. All of these

(2) Two dice were rolled once, the probability of obtaining multiple of 3 is

a. 5/9
b. 0
c. 5/6
d. ½
e. None of these

(3) Which of the following function is neither odd nor even?

a. {(1, 2), (4, 7), [illegible], (0, 4), (-4, 7)}
b. {(1, 2), (4, 7), (-1, -2), (0,0), (-4, -7)}
c. {(x, y): y = [illegible] − 1}
d. {(x, y): y = x^2 − 1}
e. f(x) = [illegible]

(4) Cube roots of unity are ________ while, fourth roots are ________

a. 1, w, w^2 ; 1, -1
b. 1, w, w^2 ; ±w, ±1
c. 1, w, w^2 ; ±i, ±1
d. 1, w, w^2 ; 1, 1
e. None of These

(5) $\lim_{n \to \infty} \left(1 + \frac{1}{n+1}\right)^n$

a. [illegible]
b. e
c. [illegible]
d. e^2 − e
e. None of these

(6) Let S = {A, B, C, D} where A = {a}, B = {a, b}, C = {a, b, c} and D = ϕ, then the missing elements of the composition table below are

U	A	B	C	D
A	A	B	C	A
B	B	B	C	B
C	C	C	C	C
D	A	[illegible]	[illegible]	?

a. C, D
b. A, A
c. A, B
d. B, B
e. C, C

(7) The middle term(s) in the expansion of $(x - 2y/3)^{10}$ is/are

a. $\frac{896}{27}x^5y^5$
b. $\frac{896}{27}x^5y^4$
c. $-\frac{896}{27}x^5y^5$
d. $\frac{-896}{27}x^5y^4$
e. None of these

(8) The T_n of the sequence 1/3, 1 2/3, 3 .... is

a. $\frac{1(4n-3)}{3}$
b. $\frac{1(4n-2)}{3}$
c. $\frac{1(4n-5)}{3}$
d. 4n − 3
e. None of these

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Dawood PT-2

(9) Any matrix of order m x n whose elements are all zero is called

a. Square Matrix
b. Rectangular Matrix
c. Triangular Matrix
d. Null Matrix
e. None of these

(10) [illegible]

a. 1
b. 1
c. 2
d. -2
e. 0

(11) $\frac{dy}{dx}$ of $y = (x^4 + 2x)^2$ is

a. [illegible]
b. [illegible]
c. [illegible]
d. [illegible]
e. Both b & d

(12) $\int \frac{dx}{x \ln 5x}$

a. 5 ln x + C
b. Ln 5x + C
c. Ln (ln x)² + C
d. Ln (ln 5x) + C
e. Ln (ln x) + C

(13) Find the area under the curve for the equation: $x^2 - 3x + 6$ intersecting line $y = x + 3$

a. 1/3 sq. units
b. 1 sq. unit
c. 1 1/3 sq. units
d. 3 1/3 sq. units
e. 2/3 sq. units

(14) A sector cut from a circle of radius 3 cm has a perimeter of 12 cm. The area of this sector is

a. 9 cm²
b. 18 cm²
c. 15 cm²
d. 25 cm²
e. None of these

(15) Find the coordinates of the point that divides the join of A(2, -5) & B(2, 1) in the ratio 3:1 internally

a. (0, 1)
b. (1, 1/3)
c. (2, -3/4)
d. (1/2, 1/3)
e. (1, 1)

(16) The center and radius of the circle having equation $2x + 6y - x^2 - y^2 = 1$ is

a. (1,3), 3
b. (3,1), 3
c. (1,3), 4
d. (3,1), 4
e. (1,3), 5

(17) A parabola has the axis at x and passes through the point (-3, 6), its equation will be

a. $y^2 = 12x$
b. $y^2 = -12x$
c. $y^2 = 3x$
d. $y^2 = -3x^2$
e. $y^2 = 12x + 72$

(18) The direction cosine of the vector formed from (1, -3, 2) and (3, -5, 1) are

a. $\frac{2}{3}, \frac{3}{3}, \frac{-6}{3}$
b. $\frac{-2}{3}, \frac{-3}{3}, \frac{6}{3}$
c. $\frac{2}{7}, \frac{-3}{7}, \frac{-6}{7}$
d. A & B both
e. None of these

(19) The equation of the straight line that passes through the point $(4, \sqrt{3})$ and makes an angle of $\pi/6$ radians with axis will be:

a. $x + \sqrt{3}y + 1 = 0$
b. $x = \sqrt{3}y - 1$
c. $x + \sqrt{3}y - 1 = 0$
d. $x - \sqrt{3}y - 1 = 0$
e. $x = \sqrt{3}y$

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(20)If the distance between (7, 1) and (3, y) is 5 units, find the value of y
a. 4
b. -2
c. A & B
d. -4
e. B & D

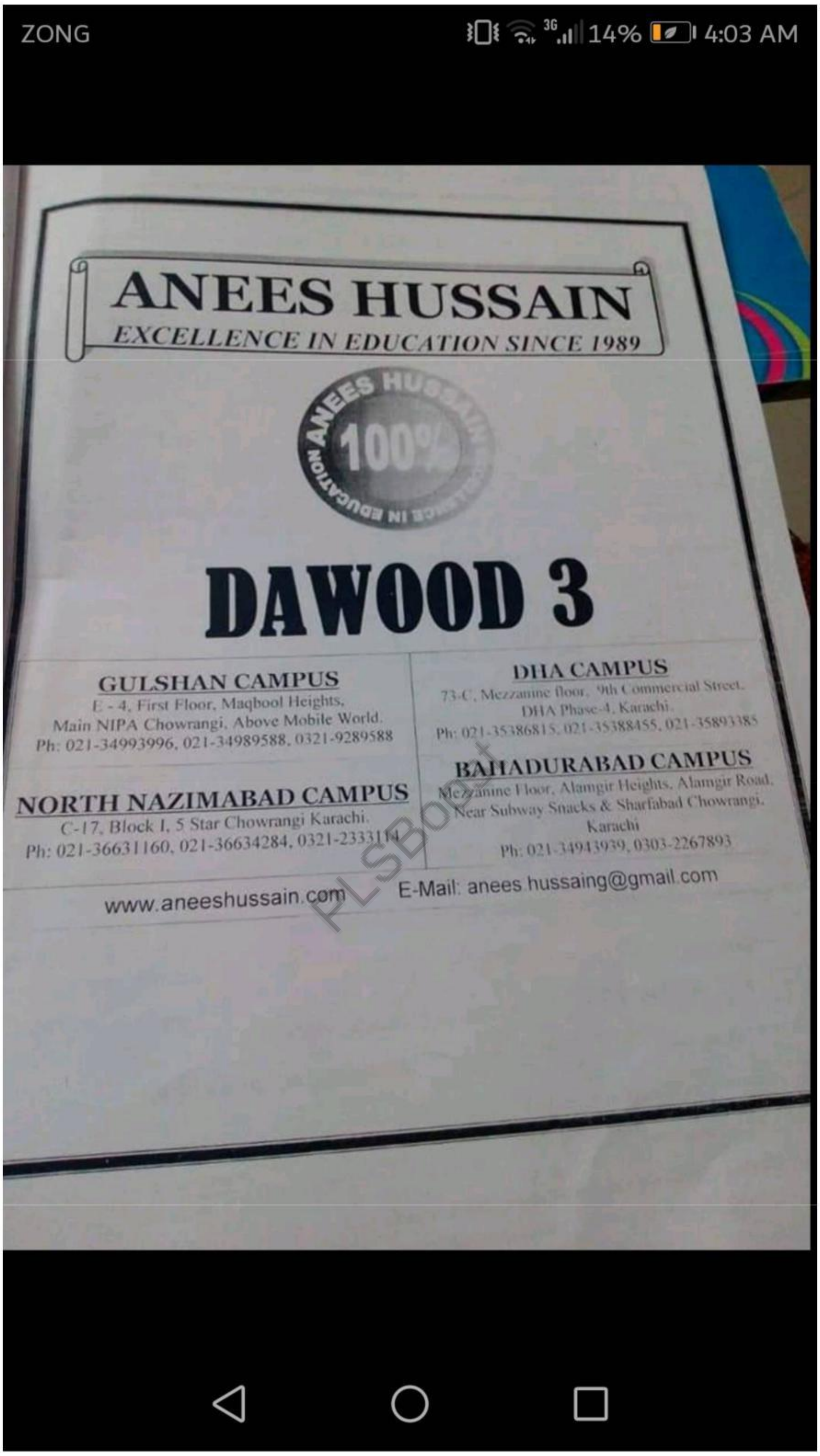
(21)If $f(x) = x^3 - 9x^2 + 15x + 3$ the relative maxima is
a. 5
b. 1
c. 10
d. -22
e. 0

(22)Let $A = \begin{vmatrix} -2 & 0 & 3 \\ 1 & -6 & 5 \\ -1 & 2 & 0 \end{vmatrix}$, the co-factor a_{31} will be
a. 1
b. 2
c. 3
d. 4
e. 5

(23)An AP is such that the sum of the first 8 term is 156 and sum of first 10 terms is 215. The fifth term will be
a. 20
b. 18.5
c. 19
d. 20.5
e. 19.5

(24)If $(1 + 5i)p - 2q = 3 + 7i$ then values of p & q will be
a. 7/5, 4/5
b. -7/5, 4/5
c. 7/5, -4/5
d. 7/5, 4/5
e. None of the these

(25)In how many ways can 4 prizes be awarded to 3 students if each student can receive all four prizes?
a. 4!
b. 3^3
c. $3^3 - 1$
d. 4
e. 3^4



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ENGLISH

Choose the correct SYNONYM for the following

1) Flotsam
   a) Wreckage b) Fade Away
   c) A float d) Line of Buoys
2) Forbearance
   a) Patience b) Animated
   c) Noisy d) Hysterical
3) Garrulous
   a) Foolish b) Sincere
   c) Playful d) Talkative
4) Glib
   a) Fluent b) Snide
   c) Satisfied d) Superficial
5) Holocaust
   a) Postulated b) Destruction
   c) Hooligan d) Grumpy

Choose the correct ANTONYM for the following

6) Fatal
   a) Deadly b) Dangerous
   c) Beneficial d) Baneful
7) Fanciful
   a) Veritable b) Whimsical
   c) Capricious d) Fantastic
8) Guilty
   a) Noted b) Childlike
   c) Childish d) Innocent
9) Gradual
   a) Regular b) Momentary
   c) Slow d) Progressive
10) Hypocrisy
   a) Cant b) Deceit
   c) Uprightness d) Pretense

Point out which underlined part of the following sentence is incorrect.

11) The (A) woman who owns (B) those cars (C) live (D) in my building.
12) Each (A) of the (B) children are (C) allowed (D) one extracurricular activity.
13) Everyone in my town (A) remember (B) the (C) night the (D) firehouse burned down.
14) The (A) cabinet are (B) empowered to take (C) this decision (D).
15) Each of the nurses were (A) scrupulously (B) careful about (C) personal cleanliness (D).

Fill in the following blanks with the most appropriate word.

16) It is ___ interesting book. It gives ___ wonderful picture of what ___ life was like in Victoria

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a) the, a, the
b) the, the, a
c) an, a, the
d) an, the, a

17) Many secretaries fail because they are not ________
a) voracious
b) versatile
c) vindictive
d) vivacious

18) Ideas are more powerful than ________ weapons.
a) legal
b) lethal
c) military
d) atomic

19) There are too many ________ and not enough serious workers.
a) dilettantes
b) sycophants
c) novices
d) zealots

20) It is easier to ________ than to accept criticism.
a) praise
b) explain
c) negotiate
d) criticize

PASSAGE

A local man, staying on the top of the floor of an old wooden house, was awakened at midnight by fire. Losing his way in the smoke-filled passage, he missed the stairway and went into another room. He picked a bundle to protect his face from fire and immediately fell through the floor below where he managed to escape through a clear doorway. The "bundle" proved to be the baby of the Mayor's wife. The "hero" was congratulated!

21) The man was called hero because
a) Expressed his willingness to risk his life to save others
b) Managed to escape from fire
c) Showed great courage in fighting the fire
d) Saved a life

22) The man went to another room because
a) He did not know where the stairway was
b) The passage was full of smoke
c) He was extremely nervous
d) He stumbled on bundle

Choose the option which is closely related to the capitalized word sentence.

23) Architect : Blue Print
a) Mason : Wall
b) Compass : Bore
c) High : Low
d) Glass : Jug

24) Argument : Debate
a) Violence : Peace
b) Fight Contest
c) Challenge : Opponent
d) Give : Take

25) Beef : Cow
a) Mutton : Buffalo
b) Sheep : Mutton
c) Mutton : Sheep
d) Pork : Cow

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CHEMISTRY

1. The process in which a solid directly converted to a gaseous state is called
   - a. Boiling
   - b. Evaporation
   - c. Sublimation
   - d. Freezing
2. The no of atoms of nitrogen represented in the formula NH_4NO_3 is
   - a. 1
   - b. 2
   - c. 3
   - d. 4
3. Which of the following technique could be used to separate the coloured from the black ink
   - a. Chromatograph
   - b. Filtration
   - c. Fractional distillation
   - d. Precipitation
4. The crystalline form of allotropes are
   - a. Chemically same
   - b. Chemically different
   - c. Some time same some time different
   - d. Depend upon substance
5. Alpha rays are
   - a. Single negative charged particles
   - b. Single positive charged particles
   - c. Double negative charged particles
   - d. Double positive charged particles
6. In an atom no two electron has the same set of quantum number
   - a. Uncertainty principle
   - b. Paulis exclusion principle
   - c. Hund's rule
   - d. Aufbau 's principle
7. Properties of a system ,which depend upon the amount of substance present in the system are called
   - a. Chemical Properties
   - b. Intensive properties
   - c. Extensive properties
   - d. Physical properties
8. The chemical formula of laughing gas is
   - a. N_2O
   - b. NO_2
   - c. N_2O_3
   - d. N_2O_4
9. The heat required to change 1 mole of iodine solid to iodine gas is called
   - a. Heat of vaporization
   - b. Heat of sublimation
   - c. Heat of fusion
   - d. Melting point
10. In the electrolysis of $CuCl_2$ the substance liberated at the anode is
    - a. Copper
    - b. Chloride
    - c. Hydrogen
    - d. Copper Chloride
11. Factor affecting the balance of a chemical equilibrium are following except
    - a. Concentration
    - b. Temperature
    - c. Enthalpy
    - d. Pressure
12. The formula for general gas equation is
    - a. $P_1V_1 = P_2V_2$
    - b. $V_1/T_1 = V_2/T_2$
    - c. $E = mC^2$
    - d. $P_1V_1/T_1 = P_2V_2/T_2$
13. The electronic configuration of the noble gas is
    - a. ns^2np^5
    - b. ns^2np^6
    - c. ns^2np^4
    - d. ns^1np^5
14. The electro negative of the element is increased
    - a. Down the group
    - b. Across the period from right to left
    - c. Across the period from left to righ
    - d. None of the above
15. Which of the following is the most suitable allotropy of sulphur
    - a. Plastic Sulphur
    - b. Monoclinic Sulphur
    - c. Rhombic Sulphur
    - d. Laughing gas
16. S-block element have the following properties except

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a. Reaction with water gives hydroxides
b. Reaction with water gives different oxides
c. Reaction with metal to gives binary ionic salts
d. Reaction with non metal to gives binary ionic salts

17. Melting and boiling of transitive elements is high because
a. Half filled d orbital
b. Metallic bonding
c. Small atomic size and valence electron
d. None of the above

18. The fat soluble vitamin is
a. Vitamin A
b. Vitamin B-12
c. Niacin
d. None of the above

19. The formula which indicates the relative number of different kinds of the ATOM in a molecules
a. Structural formula
b. Empirical Formula
c. Molecular formula
d. None of the above

20. Methane is obtained by which of the following method
a. Electrolysis of sodium acetate
b. Heating sodium acetate with soda lime
c. Electrolysis of aluminium carbide
d. All of these

21. The following reaction is an example of $C_6H_6 + Cl_2 \rightarrow heat \rightarrow C_6H_4Cl_2 + 2HCl$
a. Addition reaction
b. Reduction reaction
c. Oxidation reaction
d. Substitution reaction

22. In which of the following reaction the rate of nuleouphilic substitution reaction depend upon concentration of alkyl halide only
a. SN^1
b. SN^2
c. SN^3
d. Addition reaction

23. In the phenol
a. Alcoholic group attached on the ring
b. Hydroxyl group is attached on benzene ring
c. Alcoholic group is attached on benzene ring
d. All of the above

24. PVC prepared from
a. Ethene and hydrogen chloride
b. Ethane and hydrogen chloride
c. mathane and hydrogen chloride
d. acetylene and hydrogen chloride

25. commercially nitric acid is prepared by
a. oxidation of ammonia by bayer's method
b. oxidation of ammonia by Oswald process
c. Reduction of ammonia by Oswald process
d. oxidation of ammonia by hoop process

PHYSICS

1. The broadcasting a radio program a long distance, the effective technique of modulation is:
   - A. FM
   - B. PM
   - C. AM
   - D. BM
2. The color of light emitted by diode (LED) depends upon:
   - A. Its Biasing voltage
   - B. Its reverse voltage
   - C. Amount of Current lowing through it
   - D. Type of semi-conductor material
3. When number of loops in a closed pip increases then the wavelength:
   - A. Increases
   - B. Decreases
   - C. Remain Constant
   - D. None of these
4. Which quantity is decreases in Step-Down transformer
   - A. Voltage
   - B. Power
   - C. Current
   - D. Energy
5. The S.I. unit of measuring inductance is called
   - A. OERSTED
   - B. Gauss
   - C. Weber
   - D. Henry
6. If the potential difference across a resistor is doubled, then the rate of heat production will be.
   - A. double
   - B. Half
   - C. $\sqrt{2}$ time
   - D. Four times
7. A Ammeter is used to measure:
   - A. Emf
   - B. Current
   - C. Unknown Resistance
   - D. Internal Resistance of a cell
8. The efficiency of the Carnot's engine working between 100°C and 26°C is :
   - A. 20%
   - B. 60%
   - C. 40%
   - D. None
9. The relation for conversion between centrigrade and Fahrenheit temperatures is given by:
   - A. °F = (9/5)°C + 32
   - B. °C = (9/5)°F + 32
   - C. °F = (9/5)(C + 32)
   - D. °C = 5/9°(F − 32)
10. The image formed by an Eye piece of compound microscope is
    - A. Real and Diminished
    - B. Real and Enlarged
    - C. Virtual and Enlarged
    - D. None of these
11. The process of emitting electron by incidenting UV light on the metal sheet is
    - A. Electromagnetic
    - B. Photo electric effectS
    - C. Electrophorus
    - D. None of these
12. The center of Newton's Ring is:
    - A. Bright
    - B. Dark
    - C. Both
    - D. None
13. In Reverse Bias the diode starts behaving as a:
    - A. Conductor
    - B. Insulator
    - C. Both
    - D. None of these
14. Angular speed of daily rotation of earth in rad/hr will be:.
    - A. 1.3
    - B. 2.50

$\omega = \frac{\theta}{t} = \frac{2\pi}{24} \frac{rad}{hr}$

C. 0.26 D. 26

15. Unit of power of lens is ____
A. Meter B. Centimeter
C. Diopter D. None of these

16. Which is the equation of M_e
A. $M_e = \frac{g}{R}$ B. $M_e = \frac{G}{R}$
C. $M_e = Gg$ D. $M_e = \frac{gR^2}{G}$

17. If the force of 20N act on body at a point at axis 30° then its Ay is equal to ____
A. 0 B. 20
C. 10 D. 40

18. If the maximum height of a projectile is 40m then its maximum range will be
A. 40m B. 160m
C. 120m D. A and B both

4×40

19. The unit of Angular momentum is similar with:
A. Rydberg's Contant B. Plank's Contant
C. Coulomb's Constant D. None

20. By adding more impurities to a semiconductor the conduction property of its will
A. Increases B. Decreases
C. Remain Constant D. None of these

21. Which quantity is remains same in Step-Down transformer
A. Voltage B. Power
C. Current D. Energy

22. The colour of Black body depends on it:
A. Size B. Material
C. Temperature D. None

23. Which of the following is not affect by speed:
A. Power B. Resistance
C. Velocity D. momentum

24. which is unitless:
A. Coefficient of friction B. Refractive index
C. Magnification D. All of these

25. At the center of Earth the gravity is:
A. Maximum B. Minimum
C. Zero D. None

ANEES HUSSAIN EXCELLENCE IN EDUCATION 100%

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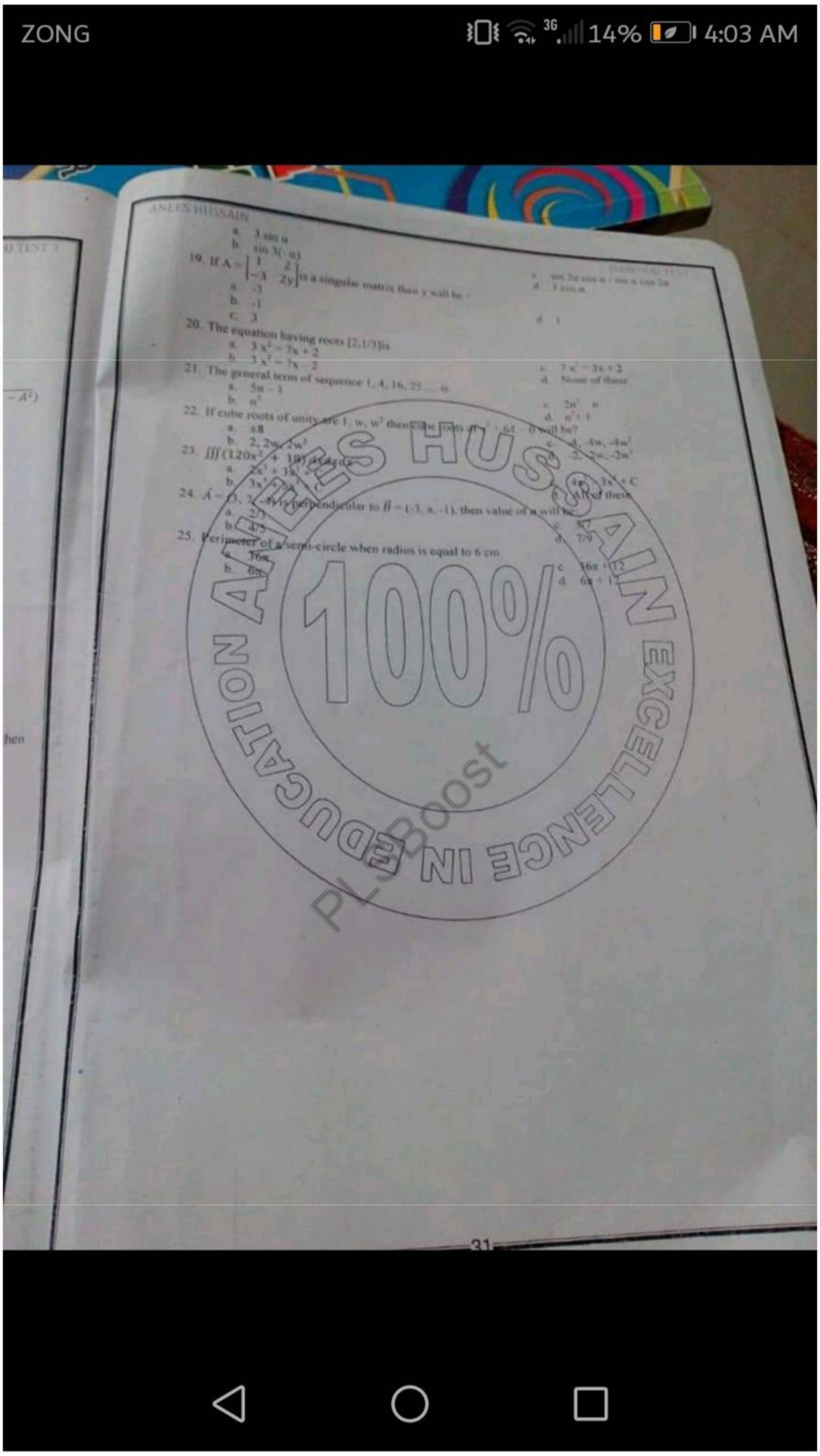
MATHEMATICS

1. $\lim_{n \to \infty} \frac{8n}{3n+2} = ?$
   a. 8
   b. ∞
   c. 8/3
   d. 3
2. Area under the curve $y = x^2 - 2x$ from $x = -1$ to $x = 3$ will be
   a. 1
   b. 1/2
   c. 1/3
   d. 4/3
3. $\sin^{-1} A + \sin^{-1} B = ?$
   a. $\sin^{-1}(A\sqrt{1-B^2} + B\sqrt{1-A^2})$
   b. $\cos^{-1}(A\sqrt{1-B^2} + B\sqrt{1-A^2})$
   c. $\sin^{-1}(A\sqrt{1-B^2} - B\sqrt{1-A^2})$
   d. None of these
4. Find the derivative of sin(sin x) w.r.t. x
   a. 0
   b. cos (cos x)
   c. cos (sin x) sin x
   d. cos (sin x) cos x
5. $\int_{-1}^{1} \ln e^x \, dx$
   a. 0
   b. 1
   c. 1/2
6. $\int e^x (\cos e^x - \sin e^x) dx$
   a. $(\cos e^x + \sin e^x)$
   b. $(\cos e^x - \sin e^x)$
   c. $e^x(\cos e^x - \sin e^x)$
   d. $e^x(\cos e^x + \sin e^x)$
7. $\frac{(x+4)^2}{9} - \frac{(y-5)^2}{?} = 1$ is a hyperbola, find its centre and vertex?
   a. (-4, 5) ; (7,-5), (1, -5)
   b. (-4, 5) ; (-7,5), (-1, 5)
   c. (4, 5) ; (-7,-5), (-1, -5)
   d. (-4, 5) ; (7,5), (1, 5)
8. Six coins are tossed simultanously, find the probability of getting at least one head?
   a. 1/64
   b. 1/8
   c. 1/32
   d. 63/64
9. $i \tanh^{-1} x = ?$
   a. $\tan^{-1} ix$
   b. $\tanh^{-1} ix$
   c. $i \tan^{-1} x$
   d. None of these
10. A car has four seats, Akbar, Amjad, Aslam and Vickey are taking their seats, if Aslam is driving then how many ways the passenger can be arranged
    a. 16
    b. 24
    c. 6
    d. 9
11. Find the equation of tangent from (1/3, 4/9) to the hyperbola $\frac{(x)^2}{4} - \frac{(y)^2}{16} = 1$?
    a. $243y = 36x + 63$
    b. $9y = 12x - 3$
    c. $27x - 9y - 5 = 0$
    d. None of these
12. If z is a complex number then express $2x - 4y + 3$ in conjugate coordinate form
    a. $(Z + \bar{Z}) - 3 = 2(Z - \bar{Z})i$
    b. $(Z - \bar{Z}) - 3 = 2(Z + \bar{Z})i$
    c. $(Z + \bar{Z}) + 3 = 2(Z - \bar{Z})i$
    d. $(Z + \bar{Z}) - 3 = 2(Z + \bar{Z})i$
13. Find the imaginary part of $z = (x + 2y)^2$
    a. 0
    b. $-4y^2$
    c. $-4xy$
    d. $x^2 - y^2$
14. Find the slope of line perpendicular to $5x - 3y + 1 = 0$
    a. -3/5
    b. 5/3
    c. -5/3
    d. 3/5
15. Find the equation of line passing through (0, 0), (0, 2) and (0, 5)?
    a. $y = 0$
    b. $x = 0$
    c. $y = x + 7$
    d. $x = y + 7$
16. Area of triangle can be calculated using:
    a. ½ base x altitude
    b. ½ (product of adjacent sides x sine of angle in between sides)
    c. Hero's formula
    d. All of these
17. If $f(x) = 2x + 5$ the $f^{-1}(3)$ will be?
    a. -2
    b. -1
    c. 0
    d. 2
18. $-\sin 3\alpha = ?$



ANEES HUSSAIN

a. 3 sin α
b. sin 3(-α)
c. sin 2α cos α - sin α cos 2α
d. 3 cos α

19. If $A = \begin{bmatrix} 1 & 2 \\ -3 & 2y \end{bmatrix}$ is a singular matrix then y will be -
a. -3
b. -1
c. 3
d. 1

20. The equation having roots {2, 1/3} is
a. $3x^2 - 7x + 2$
b. $3x^2 - 7x - 2$
c. $7x^2 - 3x + 2$
d. None of these

21. The general term of sequence 1, 4, 16, 25 ..... is
a. $5n - 1$
b. n^2
c. $2n^2 - n$
d. $n^2 + 1$

22. If cube roots of unity are 1, w, w^2 then cube roots of [illegible] 64 [illegible] will be?
a. ±8
b. 2, 2w, $2w^2$
c. 4, -4w, $-4w^2$
d. -2, -2w, $-2w^2$

23. ∭$(120x^2 + 18)\,dx\,dy\,dz$
a. $2x^5 + 3x^3$ [illegible]
b. $3x^5$ [illegible] + C
c. $4x^5 - 3x^3 + C$
d. All of these

24. $\vec{A} = (3, 2, -4)$ is perpendicular to $\vec{B} = (-3, a, -1)$, then value of a will be
a. 2/3
b. 4/5
c. 5/7
d. 7/9

25. Perimeter of a semi-circle when radius is equal to 6 cm
a. 36π
b. 6π
c. 36π + 12
d. 6π + 12

31

PLS Boost

95. The reaction $2CO + O_2 \rightarrow 2CO_2$ proceeds slower because of ______ activation energy of CO.

   A. equilibrium
   B. constant
   C. low
   D. high

96. For the first time in 1911, Henry Moseley used X-rays for the determination of:

   A. Atomic number
   B. Atomic mass
   C. Molecular mass
   D. Equivalent mass

97. The rate of a chemical reaction is directly proportional to the product of active masses of the reactants, it is referred to as:

   A. Law of conservation of energy
   B. Law of mass action
   C. Law of conservation of mass
   D. Active mass law

98. The phenomenon in which certain elements emit invisible radiations is called:

   A. Spectroscopy
   B. Radioactivity
   C. Gravimetry
   D. Chromatography

99. Real gases behave ideally at:

   A. Low temperature and low pressure
   B. Moderate temperature and low pressure
   C. High temperature and low pressure
   D. High temperature and high pressure

100. Which of the following properties depends upon the amount of matter present in the system?

   A. Density
   B. Gibb's free energy
   C. Pressure
   D. Temperature

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PHYSICS

41. Equations of kinetics are valid only when the acceleration is:
    - A. Increasing
    - B. Decreasing
    - C. Constant
    - D. Vary with time

42. Law of heat exchange is used to determine:
    - A. Coefficient of linear expansion
    - B. Coefficient of volume expansion
    - C. Ideal gas constant
    - D. Specific heat of substance

43. A solar day is the time interval between two successive appearances of the sun overhead. The time that is referred to rotation of the earth about its axis is called:
    - A. Universal Time
    - B. Length of the day
    - C. Time interval
    - D. Solar day

44. In the photo electric effect, a photon of energy $h\upsilon$ is incident on the metal surface. The energy required to eject the valance electron from the surface of the metal is expressed as:
    - A. Wave nature of the light
    - B. Compton scattering
    - C. Melting point of the metal
    - D. Work function of the metal

45. When an electron and its anti-particle positron come close enough to each other, they completely convert into radiation energy in the form of photons, this process is called:
    - A. Photo electric effect
    - B. Pair production
    - C. Annihilation of matter
    - D. Compton effect

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6. The guests broke a dozen glass at the party.

A. The guests
B. Broke
C. Glass
D. At
E. No error

Questions 7-8

Television is typical of many new scientific words which are deliberately invented from old Greek and Latin words. In this case the prefix 'tele' is Greek and means 'far' (of telephone, telegram), while the root 'vision' is derived from the Latin verb meaning 'to see'.

7. The word "Television" is invented from:

A. English and Spanish
B. French and Arabic
C. Dutch and German
D. Greek and Latin

8. Which of the following gives the meaning of the prefix "tele"?

A. near
B. close
C. far
D. away

Choose the lettered word or phrase that is most nearly opposite in meaning to the word in capital letters.

9. PATRIOTISM:

A. nationalism
B. socialism
C. obsession
D. disloyalty

10. SUPREMACY:

A. excellency
B. sovereignty
C. inferiority
D. influence

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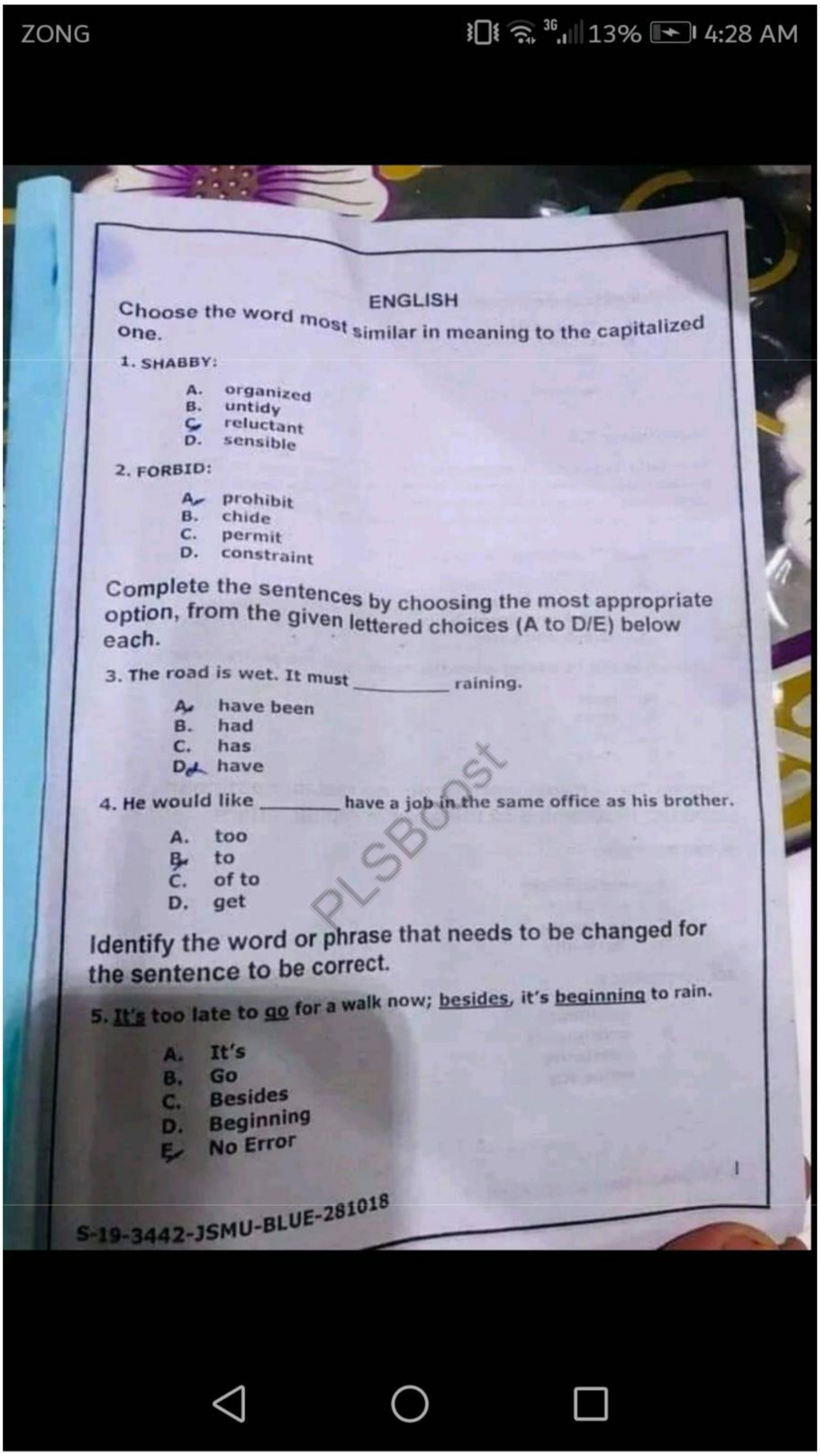

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ENGLISH

Choose the word most similar in meaning to the capitalized one.

1. SHABBY:
   - A. organized
   - B. untidy
   - C. reluctant
   - D. sensible

2. FORBID:
   - A. prohibit
   - B. chide
   - C. permit
   - D. constraint

Complete the sentences by choosing the most appropriate option, from the given lettered choices (A to D/E) below each.

3. The road is wet. It must ________ raining.
   - A. have been
   - B. had
   - C. has
   - D. have

4. He would like ________ have a job in the same office as his brother.
   - A. too
   - B. to
   - C. of to
   - D. get

Identify the word or phrase that needs to be changed for the sentence to be correct.

5. It's too late to go for a walk now; besides, it's beginning to rain.
   - A. It's
   - B. Go
   - C. Besides
   - D. Beginning
   - E. No Error

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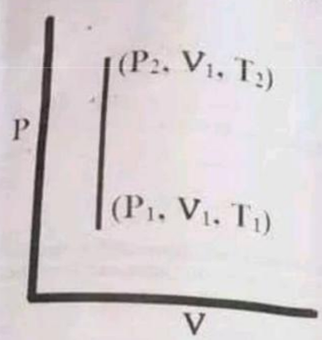
57. Which process is shown in the graph between pressure and volume?



A. Adiabatic
B. Isobaric
C. Isochoric
D. Isothermal

58. If in a circuit, 2 ampere current is drawn from the battery in 10 minutes. How much charge will flow through the circuit in this time?

A. 1200 Coulombs
B. 600 Coulombs
C. 500 Coulombs
D. 20 Coulombs

59. If the sunlight is incident on the photocell, the number of electrons emitted from the surface of the cell increases with the:

A. Increase of light frequency
B. Increase of light intensity
C. Decrease of light frequency
D. Decrease of light intensity

60. The process of natural decay of some heavy nuclides is because of the ________________ phenomenon.

A. Emission spectra
B. Nuclear fusion
C. Nuclear fission
D. Radioactivity

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61. Let X be a radioactive element having N_o nuclei. The time required to decay this element to one half of its initial number N_o is known as:

A. Law of radioactive decay
B. Decay constant
C. Half life
D. Decay Time

62. Bimetallic thermostat is an example of:

A. Electric field
B. Thermal expansion
C. Heat engine
D. Isobaric process

63. A cyclist moving towards right with an acceleration of $4m/Sec^2$. At t=0, he has travelled 5m moving towards the right at 15m/Sec. What will be his position at t=2 seconds?

A. 36
B. 38
C. 41
D. 43

64. In pure semiconductor materials, some impurities are added to change the properties of these semiconductors. This process is called:

A. Crystallography
B. Rectification
C. Intrinsic defects
D. Doping

65. Two forces equal in magnitude but opposite in direction and not acting along the same line constitute a couple. The movement of this couple will be:

A. Dependent on the location of the origin
B. Independent of the location of origin
C. Zero
D. Scalar product

66. If we increase the area of a cross section of a conductor having a length L, the resistance R of that conductor will be:

A. Zero
B. Constant
C. Increasing
D. Decreasing

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90. In electrochemical series, elements are arranged in order of their standard electrode potentials, the correct decreasing reactivity order for metals is:

A. Gold, silver, magnesium, aluminum
B. Mercury, calcium, sodium, magnesium
C. Sodium, aluminum, lead, copper
D. Potassium, silver, magnesium, aluminum

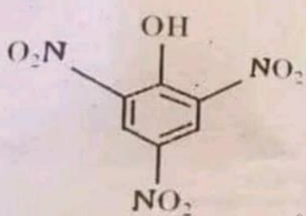
91. When electricity is passed through water in the presence of an acid oxygen ______ is produced.

A. Carbon
B. CO
C. CO_2
D. Hydrogen

92. The Neutron was discovered by:

A. Goldstein
B. Rutherford
C. J.J Thomson
D. Chadwick

93. The figure mentioned below is commonly known as:



A. Resorcinol
B. Aspirin
C. Picric acid
D. Mesitol

94. Which alkyl halide has the lowest reactivity for a particular alkyl group?

A. R-Cl
B. R-F
C. R-Br
D. R-I

46. During radioactive decay of a nucleus $_{Z}X^{A}$, β emission takes place along with daughter nucleus. Because of this beta particle emission, the mass number A of parent nucleus:

    A. Remains constant
    B. Decreases by 1
    C. Increases by 1
    D. Decreases by 2

47. In simple harmonic motion, a particle is oscillating in such a way that its total energy remains conserved. When this particle reaches its two extreme positions, its potential energy at extreme positions becomes:

    A. Minimum
    B. Maximum
    C. Zero
    D. Greater than its kinetic energy

48. In electronic circuits, a PN junction diode is used to convert alternating current into direct current. This process is called:

    A. Rectification
    B. Amplification
    C. Doping
    D. Integrated circuit

49. The process in which light bends around an obstacle is called:

    A. Diffraction of light
    B. Interference of light
    C. Reflection of light
    D. Polarization of light

50. The atomic number Z of an atom represents the:

    A. Number of protons
    B. Number of electrons and protons
    C. Number of neutrons
    D. Sum of protons and neutrons

51. If a car collides with a housefly, what will be the magnitude of the force experienced by the housefly?

    A. Much greater than the car experienced by the housefly
    B. Much lesser than the car experienced by the housefly
    C. Same as the car experienced by the housefly
    D. 10 times less than the car experienced by the housefly

67. If constant temperature conditions are applied to a gas container, then the volume of the mass of that gas tends to increase by:

- A. Decreasing the applied pressure
- B. Increasing the applied pressure
- C. Increasing applied pressure with double force
- D. Increasing the weight

68. The efficiency of the Carnot engine working between temperatures 177°C and 77°C will be:

- A. 22.22 %
- B. 23.23 %
- C. 25.25 %
- D. 30.30 %

69. Let B be the magnetic field and a conductor of length L is moved across this field. Due to this activity, a potential difference appears across the ends of the conductor. This type of potential difference is termed as:

- A. Self-Induction
- B. Mutual Induction
- C. Motional E.M.F
- D. Electrostatic potential

70. Which of the following phenomenon represents the wave nature of light?

- A. Photoelectric effect
- B. Compton scattering
- C. Interference of light
- D. Pair production

84. How many significant figures are there in 00.4793?

A. 3
B. 4
C. 5
D. 6

85. Which one of the following is NOT a meta directing group?

A. COOH
B. SO_3H
C. CHO
D. OH

86. How many chain isomers are shown by C_5H_{12}?

A. 5
B. 4
C. 3
D. 2

87. For the following exothermic reaction, decrease in temperature will shift the equilibrium towards: $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$

A. Left
B. The point of equilibrium
C. Both directions
D. Right

88. In which of the following nitrogen compounds, N has the highest oxidation state?

A. NH_2OH
B. N_2H_4
C. HNO_3
D. NH_3

89. The boiling point of water is highest than other hydrides because water molecules can form:

A. 4 hydrogen bonds
B. 3 hydrogen bonds
C. 2 hydrogen bonds
D. 1 hydrogen bonds

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77. Which of the following reactions is NOT shown by Ketones?
   - A. Reaction with HCN
   - B. Reaction with Fehling solution
   - C. Reaction with $NaHSO_3$
   - D. Reaction with 2,4-dinitrophenyl-hydrazine

78. Which one of the following is the pure carbon compound and used as reducing agent in industries?
   - A. Coal tar
   - B. Petroleum
   - C. Coal gas
   - D. Coke

79. The alloy Dura lumin is composed of ___________ of Al.
   - A. 95%
   - B. 90%
   - C. 85%
   - D. 80%

80. The normal pH of blood is:
   - A. 7.75
   - B. 7.35
   - C. 7.25
   - D. 7.05

81. The number of bond(s) between carbon and nitrogen atoms in a Nitrile is:
   - A. One sigma and one pi
   - B. Two sigma and one pi
   - C. Only sigma
   - D. One sigma and two pi

82. Methyl orange is ________ in acidic solutions.
   - A. yellow
   - B. pink
   - C. orange
   - D. red

83. The melting point of NaCl is very high 801°C, it is reduced to 600°C by addition of ___________ in Down's process.
   - A. Calcium chloride
   - B. Magnesium chloride
   - C. Aluminum chloride
   - D. Potassium chloride

CHEMISTRY

71. Which one of the following does NOT give iodoform test?

A. CH_3OH
B. CH_3CH_2OH
C. CH_3CHO
D. CH_3COCH_3

72. 106 gram of Na_2CO_3 per dm^3 of solution of Na_2CO_3 in water, the concentration of the solution will be:

A. 1N
B. 0.1M
C. 1M
D. 0.02M

73. The transparent plastic used to make combs and hair brushes is called:

A. PVA
B. PVC
C. Bakelite
D. Perspex

74. Which one of the following vitamins is required for normal growth, vision and keeping the skin healthy?

A. Vitamin D
B. Vitamin E
C. Vitamin A
D. Vitamin K

75. Zwitter ion is formed when proton goes from:

A. Amino to carboxyl group
B. Carboxyl to amino group
C. Carboxyl group only
D. Amino group only

76. Lithium and beryllium are unique in such a way that they have higher charge densities which produce strong polarizing effects due to:

A. Nonmetal
B. Solubility
C. Small size
D. Large size

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52. There are different types of waves that exist in this universe. Which of the following waves do NOT need any material medium to travel?

A. Sound waves
B. Waves produced on a string
C. Electromagnetic waves
D. Waves produced on the water surface

53. If a dielectric material is placed between two plates of a capacitor, the net capacitance of the capacitor:

A. Decreases
B. Increases
C. Remains constant
D. Zero

54. In free states, the mass of the nucleus is always:

A. Greater than the mass of its constituents
B. Equal to the mass of its constituents
C. Less than the mass of its constituents
D. Far greater than the mass of its constituents

55. If constant current is flowing through a coil then flux through that coil will become constant. In such type of arrangements, the electromotive force emf induced in coil will be:

A. Self-inductance
B. Zero
C. Mutual-inductance
D. Back emf

56. In electromagnetic spectrum of radiations, the wavelength spectrum that is visible to humans lies in the range of:

A. 100-400 nm
B. 400-700 nm
C. 700-1000 nm
D. 1000-1500 nm

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10

57. Wh
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58.

59

ADMISSION TEST # 2013

2013 Dawood UNIVERSITY OF ENGINEERING & TECHNOLOGY TEST PAPER

SECTION – I (ENGLISH)

Choose the word most similar in meaning to the capitalized one,

1. MONOTONOUS:
(a) Limited (b) Excitable (c) Extremist (d) Repeatedly
(e) Conclusive

2. PRECISE:
(a) Concise (b) Comprehensive (c) Exaggerate (d) Shortcoming (e) Explanatory

3. DESTITUTE:
(a) Disable (b) Isolated (c) Poor (d) Emangled
(e) enalize

4. EXPEDIENT
(a) Scapegoat (b) Heterogeneous (c) Suitable (d) Apathetic
(e) Concentrate

Choose the lettered word or phrase that is most nearly opposite in meaning to the world in capital letters.

5. ORTHODOX: → ancient minded
(a) Emangled (b) Unconventional (c) Conventional (d) Traditional (e) Casual

6. ECONOMICAL:
(a) Maddest (b) Efficient (c) Extravagant (d) Attentive (e) Expensive

(a) Modest (b) Efficient (c) Extravagant (d)
Attractive (e) Expensive

7. BLISS:
(a) Happiness (b) Misery (c) Comfort (d) Pleasure
(e) Stylish

8. EXHIBIT:
(a) Conceal (b) Showy (c) To sell (d) Conceive (e)
Rotated

9. BRITTLE:
(a) Easy (b) Noise (c) Tough (d) Wise

10. EMISSARY:
(a) Dictator (b) Spy (c) Merit (d) Achieve

11. JOYOUS:
(a) Genial (b) Sad (c) Gay (d) Morose

12. PLEAD:
(a) Demand (b) Justify (c) Sue (d) pray

13. VENERATE:
(a) Abominate (b) Adapt (c) Correlate (d) Involve

Identify the word or phrase that needs to be changed for the sentence to be correct:

14. The teacher gave Marta a failing grade on her composition because it was the same a composition he had already read. None
(A) (B) (C) (D) (E)

15. Your recipe for chicken is like to a recipe that my mother has. None
(A) (B) (C) (D) (E)

16. Jayne's apartment is very different from Bill's even though they are in the same building. None
(A) (B) (C) (D) (E)

17. The less one earns, the lesser one must pay in income taxes. None
(A) (B) (C) (D) (E)

FOR FREE TEST PREPARATION CONTACT
Mohammad Amin Sharif [Aptitude test Expert]
03452303294,03343537316

Read the passage to answer questions 7 – 8

Throughout history, the search for salt has played an important role in society. When it was scarce, salt was traded ounce for ounce with gold. Rome's major highway was called the Via Salaria, that is, the Salt Road. Along that road, Roman soldiers transported salt crystals from the salt flat at Ostia up the Tiber River. In return, they received a salarium or salary, which was literally money paid to soldiers to buy salt. The old saying "worth their salt", which means to be valuable, derives from the custom of payment during the Empire.

18. What does the passage mainly discuss?
(a) the old saying "worth their salt". (b) the Roman Empire
(c) salt
(d) ancient trade (e) the Roman solider

19. According to the passage, salt flats were located in:
(a) Rome (b) Ostia (c) Custom (d) Tiber
(e) Salaria

Read the passage to answer questions 7 – 8

In May 1966, the World Health Organization was authorized to initiate a global campaign to eradicate smallpox. The goal was to eliminate the disease in one decade. Because similar projects for malaria and yellow fever had failed, few believed that smallpox could actually be eradicated, but eleven years after the initial organization of the campaign, no cases were reported in the field.

20. Which of the following is the best title for the passage?
(a) The World Health Organization (b) The eradication of smallpox
(c) Smallpox vaccinations (d) Infectious diseases
(e) Vaccinations of entire villages

21. It can be inferred that:
(a) No new cases of smallpox have been reported this year
(b) Malaria and yellow fever have been eliminated
(c) Smallpox victims no longer die when they contract the disease
(d) Smallpox is not transmitted from one person to another
(e) Extensive reporting of outbreak

Complete the sentences by choosing the most appropriate word, from the choices given below each.

22. Dennis Chavez of New Mexico ______ to the House of Representatives in 1930 and to the senate in 1938.
(a) when elected (b) elected (c) who was elected (d) was elected (e) was electing

23. Knee is the joint ______ the thigh bone meets the large bone of the lower leg.
(a) where it is (b) where (c) why (d) when
(e) which

24. Quails typically have short rounded wings that enable ______ spring into full fight instantly when disturbed in ______ their hiding places.
(a) they (b) to their (c) its (d) them to
(e) in

25. When water freezes in the cracks of rocks ________ expands, causing the rocks to break apart.
(a) it (b) but (c) then (d) and (e) on

SECTION – II (MATHEMATICS) (For Pre-Engineering Group)

1. If $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$, then cofactor of 2^{nd} row are
(a) -3, 3, -11 (b) 3, -3, 11 (c) 3, 3, -11 (d) 3, 3, 11

2. If $2\Delta = a + b + c$, then the value of $\Delta = \begin{bmatrix} \Delta + c & a & b \\ c & \Delta + a & a \\ c & a & \Delta + b \end{bmatrix}$ is
(a) $4\Delta^2$ (b) $2\Delta^3$ (c) Δ^4 (d) $3\Delta^2$

3. If a, b $\in$ R then roots of $(x - a)(x - b) = abx^2$ are always
(a) real (b) dependent on a (c) not real (d) dependent on a & b both

4. $12\cdot 34\dot{5}\dot{6} =$
(a) $12\frac{3456}{9900}$ (b) $12\frac{3422}{9900}$ (c) $12\frac{122222}{9900}$ (d) B & C both

5. If a, b, c are in A.P., then $3^a, 3^b, 3^c$ are in
(a) A.P. (b) G.P. (c) H.P. (d) none of these

6. Fifteen girls compete in a race. The first three places can be taken by them in ________ ways.
(a) 12! (b) 45 (c) 15! (d) 15 x 14 x 13

7. An experiment yields 3 mutually exclusive and exhaustive events A, B, C.
If P(A) = 2P (B) = 3P(C) then, P(A) = ________
(a) $\frac{1}{11}$ (b) $\frac{2}{11}$ (c) $\frac{3}{11}$ (d) $\frac{6}{11}$

8. If the 4^{th} term in the expansion of $(px + x^{-1})^m$ is 2.5 for all x $\in$ R, then
(a) $P = \frac{5}{2}, m = 3$ (b) $P = \frac{1}{2}, m = 6$ (c) $P = \frac{1}{2}, m = 4$ (d) none of these

9. $1^3+2^3+3^3+4^3+5^3+6^3+8^3+9^3+10^3=$
(a) 55 (b) 55^2 (c) 2682 (d) none of these

10. If the elevation of the sun is 30°, then the length of the shadow cast by a tower of 150 ft height is
(a) $75\sqrt{3}\,ft$ (b) $\frac{150}{\sqrt{3}}\,ft$ (c) $150\sqrt{3}\,ft$ (d) none of these

11. The value of parameter 'a' for which the function $f(x) = 1 + ax$, $a \neq 0$ is the inverse of itself, is
(a) -2 (b) -1 (c) 1 (d) 2

12. Limit of sequence $\frac{1\cdot2}{2\cdot3}, \frac{3\cdot4}{4\cdot5}, \frac{5\cdot6}{6\cdot7}, \ldots$ ______
(a) 0 (b) 1 (c) ∞ (d) undefined

13. $\lim_{x\to\infty} \frac{4x^3+3x^2-2x}{1-2x+3x^2 4x^3} =$ ______
(a) 4 (b) 0 (c) 1 (d) -1

14. $\lim_{x\to 0} \frac{\ln(1+2x)}{3x} =$ ______
(a) does not exist (b) undefined (c) 2 (d) $\frac{2}{3}$

15. The derivation of function $f(x) = \sin^{-1}\sqrt{1-x^2}$ w.r.t $g(x) = \cos^{-1}(2x^2-1)$ is
(a) $\tan^{-1}(1+2x)$ (b) $\cos^{-1}(2x+1)$ (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

16. The velocity of a particle moving along a straight line is given by $x = 3t + t^2$. The acceleration of the particle after 4 seconds from the start is
(a) $\frac{d^2v}{dt^2}$ (b) 2 (c) 11 (d) 11 units / sec^2

17. The integral that can be evaluated is ______
(a) $\int_0^1 \frac{dx}{x-1}$ (b) $\int_0^{\frac{\pi}{2}} \sin x\, dx$ (c) $\int_1^2 \sqrt{1-x^2}\,dx$ (d) $\int_{-2}^{1} \ln x\, dx$

18. The equation of the chord of the circle $x^2+y^2-4x=0$ where mid point is $(1,0)$ is ________
(a) $y=2$ (b) $y=1$ (c) $x=2$ (d) $x=1$

19. The slope of the normal at the point $(at^2, 2at)$ of the parabola: $y^2=4ax$ is ________
(a) $\frac{1}{t}$ (b) t (c) $-t$ (d) $-\frac{1}{t}$

20. If $x+y+1=0$ touches the parabola: $y^2=\lambda x$, then λ is ________
(a) 2 (b) 4 (c) 6 (d) 8

21. The vector $2\vec{i}+4\vec{j}+7\vec{k}$ is perpendicular to the vector $2\vec{i}+6\vec{j}+x\vec{k}$ when x = ________
(a) 0 (b) 2 (c) 4 (d) 7

22. If $w^{113}+w^{23}+w^{13}-i^{n}=1$ then the least natural number n is ________
(a) 0 (b) 4 (c) 6 (d) 2

23. The measure of the two sides of a triangle of area 6 $units^{2}$ are 4 & 5 cms. Its third side is
(a) $\cdot 3$ (b) 3 units (c) 3 cms (d) none of these

24. $\int \sin\sqrt{2x}\ dx =$ ________
(a) $\sqrt{2x}\ Cos\sqrt{2x}+Sin\sqrt{2x}$
(b) $-\sqrt{2x}\ Cos\sqrt{2x}+Sin\sqrt{2x}+C$
(c) $-\sqrt{2x}\ Sin\sqrt{2x}+Cos\sqrt{2x}+C$
(d) None of these

25. The volume of the parallelopiped whose three adjacent edges are represented by the vectors : $\vec{a}=2\underline{i}-3\underline{j}+4\underline{k}, \vec{b}=\underline{i}-2\underline{j}+\underline{k}\ \&\ \vec{c}=3\underline{i}-\underline{j}+2\underline{k}$, is
(a) -7 (b) $\frac{7}{2}$ (c) $\frac{7}{3}$ (d) $\frac{7}{6}$

OR

SECTION – II (BIOLOGY) (For Pre-Medical Group)

1. Most peculiar cells of hydra are:

SECTION – III (PHYSICS)

1. A train moves on an horizontal surface with a constant velocity of 20 m/s. It requires maximum of 1500 watt power, the force retarding the motion is:
   (a) 1500 dyne (b) 15000 N (c) 515 dyne (d) 375 N
   (e) 200N

2. Body sitting in train moving at a constant velocity throws a ball straight up into the air the ball will
   (a) Behind him (b) In front of him (c) In his hand (d) Beside
   (e) Top of head

3. Newtons first law of motion is applicable to:
   (a) Inertial frames of reference (b) Relativistic frames of reference
   (c) All frames of reference (d) International frame
   (e) Smaller frames of reference

4. A spaceship orbiting the earth is an example of Newton's
   (a) 1st law (b) 2nd law (c) 3rd law (d) law of gravitation
   (e) Coulomb's law

5. When a force 4N on a mass of 20kg for a time of 2s. what is the rate of change of momentums?
   (a) 1 $kgms^2$ (b) 2$kgms^2$ (c) 4$kgms^2$ (d) 8$kgms^2$
   (e) 10k gms^{-2}

6. The S.I unit of current is:

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(a) Volt (b) Candela (c) Ampere (d) Ohm
(e) Kelvin

7. Henry is unit for
(a) Only mutual inductance (b) Only self-inductance (c) Both self & mutual inductance
(d) Magnetic flux (e) Relative inductance

8. Decibel is the unit of:
(a) Absorbing power (Absorbing co-efficient) (b) Intensity of sound (c) Loudness of sound (d) Frequency of sound (e) Velocity of sound

9. 10[illegible] seconds are equivalent to:
(a) Deci seconds (b) Nano seconds (c) Milli seconds
(d) None of these (e) Pico second

10. Which of the following is biggest unit of energy?
(a) Joule (b) Micro-Joule (c) Electron volt (d) None of these (e) B.T.U.

11. One Angstrom is equal to
(a) 10^{-8} cm (b) 10^{-8} m (c) 10^{-6} cm (d) 10^{8} mm (e) 10^{-10} cm

12. 1- Giga Hertz is equivalent to
(a) 10^{10} Hz (b) 10^{9} Hz (c) [illegible] Hz (d) 10^{-10} Hz (e) 10^{11} Hz

13. The dimensional formula for acceleration is
(a) ML^2/T (b) ML^2/T^2 (c) MLT (d) LT^{-2}
(e) LT^{-3}

14. A vector in a given direction whose magnitude is unity is called
(a) Unit vector (b) Parallel vector (c) Free vector (d) Position vector (e) Null vector

15. The x-component of the vector A making an angle of 30° with the y-axis is
(a) $A/\sqrt{2}$ (b) $\sqrt{3}/2$ (c) 0.5 A (d) O
(e) 6 A

16. If the resultant of two forces each of magnitude F is 2F then the angle between them will be
(a) 180° (b) 30° (c) 60° (d) 0°
(e) 3/2

17. Two bodies are moving in opposite direction with speed v. What is the magnitude of their relative velocity with each other?

(a) O (b) V (c) $\frac{V}{2}$ (d) 2V (e) $\frac{V^2}{2}$

18. Two forces of magnitude 20N each are acting at 30° & 60° with the x axis the y-component of the resultant force is approx:

(a) 26.32N (b) 10N (c) 20 x 0.866 N (d) 15N (e) 10 x 0.5N

19. The center of gravity may be defined as a point at which

(a) Gravitational force acts (b) The weight of the body acts (c) The body acceleration due to gravity (d) The body stops its motion (e) Nuclear force acts

20. The point at which whole weight of a body is concentrated is called

(a) Centre of mass (b) Centre of gravity (c) Origin (d) Centre of action (e) Point of rotation 2

21. The centre of gravity of a body of irregular shape lies

(a) At its centre (b) At the intersection of medians (c) At the intersection (d) At the surface of the body (e) At the interaction of vertex

22. A body or object is said to be in equilibrium when it

(a) Moves with a variable velocity (b) Moves with a uniforms velocity (c) Moves very fast in space (d) Moves very slow in space (e) move with C (speed of light)

23. A body will be in translation equilibrium if the vectors sum of external force acting on a body is

(a) Maximum (b) Minimum (c) Square (d) Zero (e) Infinite

24. A bird sets on telegraph wire 3km long. The additional tension produced in the wire is resultant

(a) Infinite (b) Equal to the weight of the bird (c) Less than the weight of the bird (d) More than the weight of the bird (e) Zero

25. The co-efficient of frictional force between two surfaces in contact does NOT depend upon

(a) The normal force passing one against the other (b) The area of surface (c) Whether the surface are stationary on relative motion (d) Whether a lubricant is used or not (e) When lubricant is heated

SECTION – IV (CHEMISTRY)

1. Heat provided to a system at constant pressure will be equal to
   (a) E b) P x V (c) H (d) P
   (e) E x V

2. The temperature 23 K corresponds to
   (a) 0°C (b) 273°C (c) 100°C (d) 50°C
   (e) 20°C

3. Volume of given mass of gas varies ________ with inverse of the temperature at constant pressure.
   (a) Inversely (b) Directly (c) Three times (d) Four times
   (e) Readily

4. The number of atoms in I molecules in $Na_2CO_3.10H_2O$.
   (a) 36 (b) 72 (c) 108 (d) 197
   e) 98

5. Which one of the following atoms A,B,C,D or E would readily form an ion with a charge of 2?

	Mass No.	Atomic No.
(a)	12	6
(b)	16	8
(c)	24	12
(d)	31	15
(e)	30	15

6. Which of the following is most likely to be the melting point of an ionic solid?
   (a) -180°C (b) -78°C (c) 114°C (d) 943°C
   e) 900°C

7. What volume does 400 cm^3 of sample of gas at 700 torr occupy when the pressure is changed to 1 atm.
   (assume temperature is constant)
   (a) 368 cm^3 (b) 184 cm^3 (c) 170 cm^3 (d) 870 cm^3 e) 970 cm

8. Dura – lumins is all of the following except
   (a) Used in making trains (b) It is composed of 95% Al + 4% C
   0.5% of Ni
   (c) Used in making aeroplanes (d) It is resistant to corrosion
   (e) It is oxides zing agent

9. Aluminum Bronze contains
   (a) 90% Cu + 10% Al (b) 90% Al + 10% CU (c) 10%
   + 10% Al + 80% Mg

ZONG 12% 4:26 AM

(d) 10% Mg + 10% Cu + 80% Al (e) 5% Mg + 10% Cu + 5% Al

10. Percentage of carbon in butanol is
(a) 50% (b) 55% (c) 60% (d) 65% (e) 80%

11. The temperature at which a solid change into liquid is called its
(a) Critical point (b) Melting point (c) Boiling point (d) None of these (e) Power point

12. Same solid when heated not melt but directly change into gaseous state it is
(a) Evaporation (b) Crystallization (c) Sublimation (d) Condensation (e) None

13. An element X forms compound having formula Na_3X_3, XO_2 and OX_3. What is X likely to be
(a) Carbon (b) Chlorine (c) Nitrogen (d) Sulphur (e) Iodine

14. Which one of the following is the molecular formula of hydrocarbon which contains 80% by mass of carbon?
(a) CH_4 (b) C_2H_4 (c) C_2H_6 (d) C_3H_6 (e) C_4H_2

15. Elements which have completely filled outermost shell and do not combine with other elements are called ____
(a) Inert elements (b) Reactive elements (c) Unstable elements (d) Rare earth elements (e) Typical

16. What is the conc of I_2 molecule in a solution containing 2.54 g of iodine in 250 cm^3 of solution?
(a) 0.01 M (b) 0.02 M (c) 0.04 M (d) 0.08 M (e) 0.2 M

17. The formula for ordinary Potash – Alum (Phitkari) is
(a) $K_2SO_4Cr_2(SO_4)_3\,24H_2O4$ g
(b) $K_2SO_4Cr(SO_4)_3$
(c) $K_2SO_4Al_2(SO_4)_3\,24H_2O$
(d) $K_2SO_4Al_2(SO_4)_4\,12H_2O$
(e) K_2SO_4

18. Nitric Acid as all the following properties except
(a) it has acidic property
(b) it has oxidizing property
(c) It has nitrating property
(d) it as bleaching property
(e) it has reducing property

19. Highest oxidation state of Nitrogen is
(a) +5 (b) +4 (c) -3 (d) +2 (e) +6

20. Alkenes are characterized by the presence of ______ bond between two carbon atoms.
(a) single (b) double (c) triple (d) none of the above
(e) non-polar

21. Alkenes are also called
(a) Parafin (b) Olefins (c) Aldehydes (d) Ketones
(e) Carboxylic Acid

22. The correct name of (IUPAC): $CH_3-CH-CH_3$ with CH_3 on the middle carbon
(a) Butane (b) 2 – mythyle propane (c) Isopropyle
(d) Dimethyle ethane (e) None of these

23. An alkane hydrocarbon chain cyclization would result in the formation of
(a) Alicyclic compound (b) Aromatic compound (c) Ethe...
(d) Cyclic compound (e) Esters

24. In 1913 Mosely found that ______ is the fundamental property of an atom.
(a) At. Number (b) At. Mass (c) At. Volume
(d) At. Charge (e) At. Structure

25. The amount of the energy required to the remove an electron from an isolated atom is called?
(a) Ionization potential (b) Ionization ability (c) Electron aff...
(d) Eledropositivity (e) Metallic character